

AUTO FAIL

☑ What causes the **AUTO FAIL light** to illuminate?

The **AUTO FAIL light** comes on when:

The active automatic pressurization controller fails or cannot properly control cabin pressure.

🔑 More detailed breakdown (line maintenance level)

The 737 has **two automatic pressurization controllers**:

- Controller **A**
- Controller **B**

Only **one is active at a time** (the other is standby, swapped each flight)

The AUTO FAIL light illuminates when:

- The **active controller loses its ability to automatically control cabin pressure**, due to:
 - Internal controller failure
 - Invalid or missing inputs
 - Inability to drive the outflow valve correctly

⚠ Typical triggers

1. Loss of critical input data

The controller needs:

- Air/ground signal
- Flight altitude
- Landing altitude
- Rate inputs

If these become invalid → controller declares a fault

2. Controller disagreement / failure

- Active controller output doesn't match expected behavior
- Internal fault detection trips

3. Outflow valve control problem

- Controller commands movement, but:
 - Valve doesn't respond correctly
- Feedback mismatch can trigger AUTO FAIL

What happens operationally?

When **AUTO FAIL** comes on:

- The system **automatically switches to ALTN (alternate controller)**
- You'll typically see:
 - **AUTO FAIL (amber)**
 - **ALTN (green)** illuminate (as shown in your image)

 This indicates:

- Primary auto mode failed
- Alternate auto mode is now controlling pressurization

Important distinction

Light	Meaning
AUTO FAIL	Active auto controller failed
ALTN (green)	Backup controller is now in control
MANUAL	Manual mode selected or required if both fail

When it becomes a bigger problem

If you also get:

- **Both auto systems fail**
- Or ALTN cannot control properly

→ You must go to **MANUAL pressurization control**

Quick AMT takeaway

If you see AUTO FAIL:

- Think:

- **Active press controller fault (A or B depending on selection)**
- Confirm:
 - System switched to ALTN
- Possible write-ups:
 - Pressurization controller fault
 - Input/feedback signal issue
 - Outflow valve control issue

OFF SCHED DESCEND

OFF SCHED DESCENT light — what it means

The **OFF SCHED DESCENT (amber)** light illuminates when:

The cabin altitude is higher than what the pressurization system expects for that phase of flight.

In simple terms: 🖐️ The airplane thinks the cabin is **not pressurizing properly**

🖐️ Or it is **climbing faster than planned**

How the system determines this

The pressurization controller compares:

- **Actual cabin altitude**
- **Programmed cabin profile** (based on:
 - Flight altitude (FLT ALT)
 - Landing altitude (LAND ALT)
 - Ground/air logic)

If:

Cabin altitude is **too high compared to the scheduled profile**

 **OFF SCHED DESCENT light comes on**

⚠ Common causes (line maintenance mindset)

1. **Loss of bleed air / insufficient pressurization source**

- Engine bleed issue (PRSOV, bleed trip, etc.)
- APU bleed issues
- Packs not supplying enough air

👉 Cabin begins to climb → triggers light

2. **Outflow valve open too much**

- Valve stuck open or leaking
- Controller not positioning valve correctly

👉 Air escapes faster than it should

3. **Pressurization controller problem**

- Improper schedule calculation
- Wrong inputs (FLT ALT / LAND ALT mis-set)

4. **High aircraft climb rate**

- Airplane climbs faster than pressurization system can keep up

👉 Temporary OFF SCHED DESCENT can occur

5. **Door or fuselage leak**

- Poor seal or structural leak
- Less common but important

✈ What you'll typically see with it

In the exact panel you showed:

- **OFF SCHED DESCENT (amber)** illuminated
- Possibly:
 - Increasing **cabin altitude**
 - Higher than normal **cabin climb rate**
- If it worsens:

- May lead to **cabin altitude warning horn**

Key difference vs AUTO FAIL

Light	Meaning
AUTO FAIL	Controller failed
OFF SCHED DESCENT	System is working, but cabin isn't following the schedule

 So:

- AUTO FAIL = **brain problem**
- OFF SCHED DESCENT = **performance/regulation problem**

Quick troubleshooting logic (AMT view)

If you see OFF SCHED DESCENT:

1. Check:
 - a. Bleed air status
 - b. Pack operation
2. Look at:
 - a. Outflow valve position
3. Verify:
 - a. FLT ALT / LAND ALT set correctly
4. Consider:
 - a. Possible leaks or valve malfunction

Real-world shortcut

Think of it like this:

"The airplane expected the cabin to be lower... but it isn't."

If you want, I can tie this directly into **PRSOV failures** and show how a bleed problem can trigger OFF SCHED DESCENT—that's a really common write-up path.

ALTN LIGHT

☑ **ALTN Light (Green) — What it means on the 737**

On the panel in your image, the **ALTN light illuminated (green)** indicates:

The alternate automatic pressurization controller is currently in control of the system.

🔗 How it works (simple system logic)

The 737 has:

- **Two automatic pressurization controllers**
 - **AUTO (primary – A or B depending on selection)**
 - **ALTN (backup controller)**

Only **one controller runs at a time**.

💡 When the ALTN light comes on

☑ **Normal reason (most common):**

- The **active AUTO controller fails**
- System **automatically switches to the alternate controller**

➡ You will see:

- **AUTO FAIL (amber)**
- **ALTN (green)**

👉 This is exactly what your image shows

☑ **Other possible reason:**

- Manual selection of alternate controller (less common in line ops)

✈ What it means operationally

- Pressurization is still **fully automatic**
- The airplane is **still controlling cabin pressure normally**

- You have **lost redundancy**, but not function

⚠ Important mindset (AMT / troubleshooting)

ALTN light ON by itself is NOT a failure

It only tells you:

“The backup system is doing the job now.”

MANUAL

☑ **MANUAL Light (Green) — What it means on the 737**

The **MANUAL light illuminated (green)** indicates:

The pressurization system is being controlled manually (not by the automatic controllers).

🔗 What triggers the MANUAL light

The MANUAL light comes on when:

1. **Manual mode is selected**

- The pressurization mode selector is moved to **MAN**
- The crew is now directly controlling the **outflow valve** using the manual switch

2. **Both automatic controllers are not available (failure scenario)**

- Primary AUTO fails → AUTO FAIL
- Alternate AUTO fails or unusable
- Crew selects MANUAL to maintain control

✂ What changes in MANUAL mode

When MANUAL is active:

- Automatic system is **completely bypassed**

- No programmed cabin schedule
- No automatic altitude/rate control

👉 Instead:

- Crew uses the **manual valve switch** to:
 - OPEN → increase cabin altitude (depressurize)
 - CLOSE → decrease cabin altitude (pressurize)

⚠️ Important characteristics

- The system becomes **purely manual control**
- Much higher workload
- Requires continuous monitoring of:
 - Cabin altitude
 - Cabin rate

🔗 Relationship to other lights

Lights	Meaning
MANUAL (green)	Manual mode active
AUTO FAIL + ALTN	Backup auto working normally ✅
AUTO FAIL + no ALTN → MANUAL used	Both autos unavailable ⚠️
OFF SCHED DESCENT + MANUAL	Crew controlling because auto couldn't maintain pressure

🧠 Easy way to think about it

- **AUTO** = system flying pressurization
- **ALTN** = backup system flying
- **MANUAL** = **you are flying the outflow valve yourself**

🔍 Line maintenance perspective

If you see a write-up involving MANUAL mode:

Possible causes:

- Both pressurization controllers faulty
- Controller switching issue

- Outflow valve auto control failure (but manual still works)

👉 Key diagnostic question:

Did manual mode **restore normal control**?

- ✅ If YES → controllers likely at fault
- ❌ If NO → suspect **outflow valve or mechanical issue**

💡 Real-world troubleshooting tip

If:

- AUTO mode = unstable pressurization
- ALTN = also bad
- MANUAL = stable when crew controls valve

👉 Strong indication:

Controller problem, not the valve

⚡ Summary

MANUAL light = crew is directly controlling cabin pressure because automatic control is not being used or not available

PACK LIGHT

☑ **PACK Light (Amber) — What it means on the 737**

Looking at your image, the **PACK lights (amber) on both L and R sides** are illuminated inside the bleed trip/overheat boxes.

🔔 **Definition**

The **PACK light (amber)** illuminates when the **air conditioning pack has automatically shut down**.

🔑 **Why a pack shuts down (what triggers it)**

The PACK light comes on when the system detects a condition that requires **protecting the pack**:

1. 💧 **Pack Overheat**

- The pack temperature gets too high
- Causes:
 - Cooling airflow problem
 - Pack internal fault
- Result:
 - Pack automatically **shuts down**
 - **PACK light comes on**

2. 🔑 **Bleed air supply problem**

- Insufficient or interrupted bleed air:
 - PRSOV issue
 - Bleed trip off
 - Isolation valve configuration problem

👉 Pack cannot operate → shuts down

3. ⚠️ **Associated duct overheat or bleed trip**

From your panel:

- You also have:
 - **WING-BODY OVERHEAT (amber)**
 - **BLEED TRIP OFF (amber)**

👉 These are directly tied together

Chain reaction example:

- High duct temp or pressure →
- BLEED TRIP OFF →
- Pack loses bleed source →
- **PACK light illuminates**

4. ❄️ **Low airflow / cooling issue**

- Cooling air (ram air) insufficient
- Ground ops, high temp conditions
- Pack cannot maintain proper operation

✈️ What happens when PACK light is on

- That pack is **NOT producing conditioned air**
- Effects:
 - Reduced air conditioning
 - Reduced pressurization capability

👉 If both packs are off:

- Pressurization performance is degraded significantly
- Can lead to:
 - **OFF SCHED DESCENT**
 - Rising cabin altitude

🔑 How it ties to your earlier questions

This image shows a **classic bleed/pack problem scenario**:

You have:

- **PACK lights ON**
- **WING-BODY OVERHEAT**
- **BLEED TRIP OFF**

👉 That tells you:

The packs didn't fail first — they were **shut down as a result of a bleed air system problem**

🧠 Quick troubleshooting logic (AMT mindset)

If PACK light is on:

Step 1:

Check:

- **BLEED TRIP OFF?**
- **OVERHEAT indication?**

👉 If YES → problem is upstream (bleed system)

Step 2:

Check:

- Bleed valves (PRSOV, etc.)
- Duct pressure

WING TO BODY OVERHEAT

WING-BODY OVERHEAT light (Amber) — What it means on the 737

Looking at your panel, both **WING-BODY OVERHEAT lights are illuminated** (left and right). This is a **bleed air duct temperature problem**, not a pack itself.

Definition

The **WING-BODY OVERHEAT light** illuminates when **bleed air duct temperature becomes too high** in the wing or body duct area.

What system is involved

This monitors the **bleed air ducts**:

- From engine/APU bleed source
- Through the **wing leading edge area**
- Into the **fuselage (body ducts)**

These ducts carry **hot compressed air**, so overheating is critical.

What causes WING-BODY OVERHEAT

1. **Bleed air overtemperature**

- PRSOV failing to regulate properly (too much pressure/temperature)
- Engine bleed air too hot
- High power/high load conditions

2. **Cooling issue**

- Insufficient airflow around ducts (especially on ground)
- Poor ventilation

3. 📖 **Leak in bleed duct**

- Crack, loose coupling, or seal failure

👉 This is a **big one**

- Hot bleed air escapes into surrounding structure
- Temperature rises rapidly
- Sensors trigger overheat

4. ⚠️ **Sensor detection (loop system)**

- Overheat detection loops sense high temperature
- If threshold exceeded → light illuminates

📋 **What happens next (automatic protection)**

When WING-BODY OVERHEAT occurs:

1. ✅ **BLEED TRIP OFF light illuminates**
2. ✅ Bleed valve (PRSOV) is commanded closed
3. ✅ **Downstream air supply is cut**
4. ✅ **PACKS shut down → PACK lights come on**

🔗 **Tie this to YOUR PANEL (very important)**

You have all of these illuminated:

- **WING-BODY OVERHEAT (amber)**
- **BLEED TRIP OFF (amber)**
- **PACK (amber)**

👉 This is a **perfect textbook chain**:

Overheat detected → bleed trips OFF → packs lose air → packs shut down

🧠 **Key interpretation (line maintenance)**

If you see WING-BODY OVERHEAT:

⚠️ **Think “duct problem first” — NOT pack problem**

🔍 Troubleshooting logic (AMT mindset)

Step 1: Confirm

- Is it left, right, or both sides?
 - **Both sides (like your panel)** → suspect:
 - Systemic issue (high bleed pressure/temp)
 - Not likely just a single duct leak

Step 2: Reset attempt

- Use **TRIP RESET**
- Monitor:
 - Does it trip again immediately?

👉 If YES:

- Active overheat still present → real fault

Step 3: Likely causes to investigate

- **PRSOV malfunction (fail open / overpressure)**
- Bleed air temperature control issue
- Actual duct leak (especially if one side only)

⚡ Quick field insight

Single side overheat:

👉 Likely **duct leak or local issue**

Both sides overheat (like your image):

👉 Think:

- **Control problem (PRSOV, regulation issue)**
- **Overpressure / overtemp affecting entire system**

💡 Simple way to remember

WING-BODY OVERHEAT = the air is too hot where it shouldn't be

☑ Summary

- **WING-BODY OVERHEAT (amber)** = Bleed duct temperature too high
- Causes:
 - Overtemp bleed air
 - Duct leak
 - Regulation failure
- Triggers:
 - **BLEED TRIP OFF**
 - **PACK shutdown**

BLEED TRIP OFF

💡 **BLEED TRIP OFF Light (Amber) — What it means on the 737**

On the panel you provided, both **L and R BLEED TRIP OFF lights are illuminated**, along with:

- **WING-BODY OVERHEAT**
- **PACK lights**

That combination tells a very clear story.

☑ Definition

The **BLEED TRIP OFF light (amber)** means the **bleed air valve (PRSOV) has automatically closed due to a protection trip**.

🔑 **What actually “tripped”**

The system has protection for:

- **Overpressure**
- **Overtemperature (overheat)**

When either occurs: 🙌 The system **commands the PRSOV closed** 🙌 Bleed air flow is **shut off** 🙌
BLEED TRIP OFF light comes on

⚠️ What causes BLEED TRIP OFF

1. 💧 Duct Overtemperature (most common with your panel)

- Detected by **wing-body overheat sensors**
- You already have:
 - ☒ **WING-BODY OVERHEAT lights ON**

👉 This is a **direct trigger**

2. 📄 Overpressure

- Duct pressure exceeds limit (~60–65 psi range)
- Often caused by:
 - **PRSOV fail open (not regulating)**

👉 This is where your earlier question ties in directly

3. ⚙️ Valve/control malfunction

- PRSOV not responding correctly
- System trips as a protective measure

📄 What happens after the trip

When BLEED TRIP OFF activates:

1. ☒ **PRSOV closes**
2. ☒ Bleed air supply is cut
3. ☒ **PACKS shut down → PACK lights come ON**
4. ☒ Pressurization capability reduced

🔗 Tie ALL your lights together (this is key)

Your panel shows:

- **WING-BODY OVERHEAT**
- **BLEED TRIP OFF**
- **PACK lights**

👉 This is a **textbook chain reaction**:

Bleed air too hot or too high pressure



WING-BODY OVERHEAT sensed



System trips PRSOV closed



BLEED TRIP OFF light



Packs lose bleed air



PACK lights ON

Line maintenance interpretation

If you see BLEED TRIP OFF:

👉 Ask yourself:

- **Did overheat happen?**
 -  YES (in your case)

→ Root cause likely:

- Overtemp condition
- Possibly PRSOV regulation problem

Reset function

You have the **TRIP RESET button** on the panel.

What it does:

- Attempts to **reopen the bleed valve**

What to watch for:

-  **Reset works and stays normal** → Possible transient condition
-  **Trips again immediately** → Active fault:
 - Duct still too hot
 - Overpressure still present
 - PRSOV malfunction

🔑 Important distinction

Light	Meaning
BLEED TRIP OFF	Valve was forced closed by protection logic
BLEED OFF (	Crew manually turned bleed off

👉 "TRIP" = automatic protection

👉 "OFF" = manual selection

⚡ Real-world shortcut

BLEED TRIP OFF = the system protected itself from damaging conditions

☑ Summary

- **BLEED TRIP OFF (amber)** = PRSOV automatically shut due to:
 - Overheat ✅
 - Overpressure ✅
- Leads to:
 - Loss of bleed air
 - Pack shutdown
- In your image: 👉 Root issue is **overheat condition (likely driven by bleed regulation problem or duct issue)**

💡 Big-picture takeaway

With **ALL your lights together**, the most likely scenario is:

⚠ **Bleed air problem (possibly PRSOV over-regulation or duct issue) caused overheat → triggered trip → shut down packs**

ZONE TEMP

☑ **ZONE TEMP Light (Amber)** — What it means on the 737

In your image, all three **ZONE TEMP lights (amber)** are illuminated:

- **CONT CAB (Control Cabin)**
- **FWD CAB**
- **AFT CAB**

Definition

The **ZONE TEMP light** comes on when the system detects a **temperature control problem in that specific cabin zone**.

What the system is doing

Each cabin zone has:

- A **temperature controller**
- A **trim air system** (mixes hot bleed air with conditioned air)
- Temperature sensors

The system tries to maintain whatever is selected on the temp knobs.

What triggers a ZONE TEMP light

1. **Overtemperature in a zone**

- Too much hot trim air
- Temperature exceeds limit

 System:

- Shuts off trim air to that zone
- Turns on **ZONE TEMP light**

2. **Control failure (can't regulate properly)**

- Controller not maintaining selected temp
- Sensor mismatch

3. **Trim air system issue**

- Trim air valve stuck open/closed
- Loss of trim air supply

4. ⚙️ Temperature controller failure

- Zone temp control channel fails internally

🔑 KEY insight from YOUR panel (very important)

All three ZONE TEMP lights are ON **simultaneously**.

👉 That is NOT normal for individual zone problems.

This strongly indicates:

⚠️ **A system-wide issue — not three separate zone failures**

💧 Tie it to the rest of your panel

From your earlier images, you also have:

- **BLEED TRIP OFF**
- **WING-BODY OVERHEAT**
- **PACK lights ON**

🔗 Here's what likely happened:

Bleed air problem (overheat or overpressure)

↓

Bleed system trips OFF

↓

Packs shut down (no conditioned air)

↓

No proper air supply to cabin

↓

Temperature control system can't regulate

↓

ZONE TEMP lights illuminate

🗣️ Line maintenance interpretation

If ONE ZONE TEMP light:

👉 Think:

- Local issue (sensor, valve, controller)

If ALL ZONE TEMP lights (like your panel):

👉 Think:

- **Upstream system problem:**
 - Bleed air
 - Pack operation
 - Trim air supply failure

✅ Not likely:

- Three controllers failing at once

🔗 **What the system is doing when light comes on**

- Turns off **trim air to that zone**
- Prevents overheating
- Leaves zone temperature less controlled

✈ **Operational effect**

- Cabin may become:
 - Too cold (no trim hot air)
 - Uneven temperature distribution

⚡ **Simple way to remember**

ZONE TEMP = "I can't control the temperature in this zone"

☑ **Summary**

- **ZONE TEMP (amber)** = Temperature control problem in that zone
- Causes:
 - Overtemp
 - Trim air failure
 - Controller fault
- In YOUR image: 👉 All zones = **system-wide issue (bleed/pack failure), not individual zones**

💡 Big-picture takeaway (this whole panel)

Your images together show a **classic cascade failure**:

Bleed system problem → Overheat → BLEED TRIP → Packs shut down → Temp control lost → ZONE TEMP lights

DUAL BLEED

⚠️ **DUAL BLEED Light (Amber) — What it means on the 737**

On your panel (bottom of the air conditioning section), the **DUAL BLEED light (amber)** is illuminated.

🔔 **Definition**

The **DUAL BLEED light** comes on when the system detects **both the APU bleed air AND an engine bleed air source supplying the manifold at the same time**.

🔧 **Why that's a problem**

The 737 is designed to **avoid having two high-pressure bleed sources feeding the same duct simultaneously**, because:

- It can cause:
 - **Overpressure**
 - **Overtemperature**
 - **Duct damage risk**

👉 So the system protects itself.

⚠️ **What triggers the DUAL BLEED light**

1. ☒ **APU bleed ON + Engine bleed ON (same side or manifold)**

- Most common cause
- Especially:

- On the ground during engine start or pack operation

2. **Isolation valve configuration**

- If the **isolation valve is OPEN**:
 - APU air and engine air can mix

👉 System detects this → triggers light

3. **Improper configuration**

- Pack switches, bleed switches, and isolation valve not set correctly

What the system does automatically

When DUAL BLEED is detected:

👉 The system **automatically closes the isolation valve**

Purpose:

- Separate the two bleed sources
- Prevent mixing

Tie it to YOUR panel (very important)

You previously showed:

- **BLEED TRIP OFF**
- **WING-BODY OVERHEAT**
- **PACK lights**
- **ZONE TEMP lights**

Now add:

- **DUAL BLEED**

👉 This suggests a bigger picture:

You may have had:

APU bleed ON + engine bleed ON



Isolation valve open or misconfigured



DUAL BLEED condition



Overpressure / overtemperature risk



WING-BODY OVERHEAT



BLEED TRIP OFF



PACK shutdown



ZONE TEMP lights

Line maintenance interpretation

If you see DUAL BLEED:

Step 1:

Check:

- APU bleed switch
- Engine bleed switches
- Isolation valve position

Step 2:

Ask:

“Are both bleed sources feeding the same manifold?”

Step 3:

Correct configuration:

- Either:
 - Use APU bleed alone
 - Or engine bleeds alone
- Ensure isolation valve logic is correct

Important distinction

Light	Meaning
DUAL BLEED	Two bleed sources feeding system at once 
BLEED TRIP OFF	System shut valve due to fault
WING-BODY OVERHEAT	Temperature too high in ducts

Simple way to remember

DUAL BLEED = “Too many air sources feeding the same system”

Summary

- **DUAL BLEED (amber)** = APU + engine bleed supplying system simultaneously
- System response:
 - Closes isolation valve automatically
- Risk if not corrected:
 - Overpressure / overheat (which matches your panel)

Big-picture takeaway (your whole scenario)

With **everything you showed**, the most likely root chain is:

 **Improper bleed configuration (dual bleed) → overheat → bleed trip → pack loss → temp control failure**

RAM DOOR FULL OPEN

RAM DOOR FULL OPEN Light (Blue) — What it means on the 737

Looking at your panel, the **RAM DOOR FULL OPEN lights (blue)** are illuminated for both packs.

Definition

The **RAM DOOR FULL OPEN light** indicates that the **ram air door for that pack is fully open**.

What the ram air door does

Each pack uses a **ram air system** for cooling:

- Outside air enters through the **ram air inlet door**
- Passes through the **pack heat exchangers**
- Removes heat from the bleed air

 It's essential for **pack cooling**

When this light is NORMAL

1. In flight

- Doors are **modulated or fully open**
- Provides cooling airflow

2. On the ground with packs operating

- Especially in:
 - Warm weather
 - High pack load

 Doors often go **full open** to maximize cooling

3. When packs are commanded but need maximum cooling

- System automatically opens doors fully

When it becomes abnormal

The light itself **is not a fault indicator**—it's a **status light**.

But it can point to problems if:

1. Door stuck full open

- Doesn't modulate

- Could cause:
 - Overcooling or inefficiency

2. 🛠️ **Control problem**

- Actuator or controller issue

3. 🚫 **Unexpected configuration**

- Door full open when packs are OFF (shouldn't normally happen)

🔗 **Tie it to YOUR panel scenario**

Earlier, your system showed:

- **PACK lights ON (packs shut down)**
- **BLEED TRIP OFF**
- **WING-BODY OVERHEAT**
- **ZONE TEMP lights**

Now:

- **RAM DOOR FULL OPEN (blue)**

👉 What this likely means in your case:

Even though the packs are **NOT operating**, the system has driven:

Ram air doors FULL OPEN

Why?

- To:
 - **Maximize cooling**
 - Help remove residual heat from the system
 - Respond to prior overheat condition

🧠 **Key maintenance insight**

If you see:

- **RAM DOOR FULL OPEN + PACK lights ON**

👉 Ask:

"Are the packs actually running?"

- If **NO (like your panel)**:
 - Doors are likely open due to:
 - Overheat recovery
 - Control response to previous condition

🔑 Simple comparison

Light	Meaning
RAM DOOR FULL OPEN (blue)	Cooling door fully open (status)
PACK (amber)	Pack shut down
WING-BODY OVERHEAT	Too hot in ducts
BLEED TRIP OFF	Bleed shut down

⚡ Simple way to remember

RAM DOOR FULL OPEN = maximum cooling airflow available

☑ Summary

- **Blue light = NOT a failure**
- Indicates:
 - Ram air door is fully open
- Can be:
 - Normal (cooling demand high)
 - Or part of system response after a fault

💡 Big-picture (your entire panel decoded)

Putting EVERYTHING together:

⚠ Bleed system problem → overheat → bleed trip → packs shut down → temp control lost → ram air doors open to cool system

OVERHEAT

HYD OVERHEAT Light (Amber) — What it means on the 737

Looking at your image, you have:

- **OVERHEAT (amber)** lights illuminated for hydraulic systems
- Along with **LOW PRESSURE (amber)** lights

This is a **serious hydraulic system warning combination**.

Definition

The **HYD OVERHEAT light** indicates that the **hydraulic fluid temperature in that system is too high**.

737 Hydraulic system quick context

The 737 has **two main hydraulic systems**:

- **System A**
- **System B**

Each system has:

- Fluid reservoir
- Engine-driven pump
- Electric pump
- Cooling (via fuel heat exchangers)

What causes HYD OVERHEAT

1. **Loss of cooling (very common)**

- Hydraulic fluid is cooled by:
 - **Fuel (fuel-to-hydraulic heat exchanger)**

 If:

- Low fuel
- Poor fuel flow
- Heat exchanger issue

→ Fluid overheats

2. **Pump operating under abnormal conditions**

- Pump running **dry or cavitating**
- Pump working too hard (restriction or blockage)

3. **Low fluid quantity**

- Less fluid = heats up faster
- Often tied to:
 - Leaks

4. **System leak**

- Causes:
 - Low pressure
 - Heat buildup
- Fits your panel (OVERHEAT + LOW PRESSURE together)

VERY IMPORTANT: Your panel combination

You have:

-  **HYD OVERHEAT**
-  **LOW PRESSURE**
-  Multiple pumps selected ON

 This combination strongly suggests:

 **Hydraulic system failure or severe degradation (likely fluid loss or pump issue)**

What LOW PRESSURE adds to the story

When you see:

- **OVERHEAT ONLY**
 - Cooling issue likely
- **LOW PRESSURE ONLY**
 - Pump not supplying pressure
- **OVERHEAT + LOW PRESSURE (like yours)**
 -  **Major issue:**

- Pump running but not building pressure
- Fluid likely low or system compromised

Line maintenance troubleshooting mindset

Step 1: Identify system

- System A or B (or both)

Step 2: Think causes

If BOTH lights:

- Check:
 - Fluid quantity
 - Leaks
 - Pump condition
 - System integrity

Step 3: Likely root causes

- Hydraulic leak 
- Pump cavitation 
- Low reservoir quantity 
- Overworked pump with no output 

Operational effect

Hydraulic systems power critical components:

- Flight controls
- Landing gear
- Brakes
- Nose wheel steering

 Depending on system affected:

- You may lose:
 - Some control redundancy
 - Certain aircraft functions

Important interpretation

OVERHEAT = fluid too hot

LOW PRESSURE = not enough hydraulic pressure

 Together:

System is failing AND overheating while doing it

Simple way to remember

 +  = "It's hot and it's not working"

Summary

- **HYD OVERHEAT** = hydraulic fluid temperature too high
- Causes:
 - Loss of cooling
 - Low fluid
 - Pump issues
- With **LOW PRESSURE (like your image)**:  Indicates **serious system problem (likely fluid loss or pump failure)**

Big-picture (your overall systems scenario)

Across all your panels, you've basically built a **full aircraft systems cascade scenario**:

- Bleed system failure 
- Pack shutdown 
- Temperature control loss 
- Hydraulic degradation 

 This is exactly the kind of **multi-system failure picture** you'd analyze step-by-step on the line

LOW PRESSURE

LOW PRESSURE Light (Amber) — What it means on the 737 (Hydraulics)

Looking at your panel, there are multiple **LOW PRESSURE lights illuminated** above the hydraulic pump switches (ENG and ELEC pumps for Systems A and B).

Definition

The **LOW PRESSURE light** indicates that the **hydraulic pump is not producing adequate pressure**.

What “low pressure” really means

Each pump (engine-driven or electric) is supposed to supply about:

- **~3000 psi**

If pressure drops below a set threshold:  The **LOW PRESSURE light comes ON**

What causes LOW PRESSURE

1. **Pump not operating**

- Pump switch OFF
- Pump failed internally
- No mechanical drive (engine pump)

2. **Low hydraulic fluid**

- Not enough fluid in reservoir
- Pump can't build pressure

 Very common when you also see:

- **HYD OVERHEAT** (like your panel)

3. **Hydraulic leak**

- Fluid escaping system
- Pressure drops

 Strongly suspected when:

- Multiple LOW PRESSURE lights

- System A or B degraded

4. **Pump cavitation / air in system**

- Pump running but drawing air instead of fluid
- Results:
 - Low pressure
 - Overheating

5. **Electric pump issue**

- Electrical failure
- Motor not running

CRITICAL: Your panel combination

You have:

-  **LOW PRESSURE**
-  **HYD OVERHEAT**
-  Both ENG and ELEC pump lights affected

 This is NOT a simple pump OFF situation

This strongly indicates:

 **System is being commanded ON but cannot produce pressure**

Interpretation (AMT mindset)

If:

- Pump OFF → LOW PRESSURE
 Normal indication

If:

- Pump ON + LOW PRESSURE (your case)
 Problem exists:
- Fluid loss 

- Pump failure ✓
- System leak ✓

💧 Tie **LOW PRESSURE** + **OVERHEAT** together

This combination is key:

Low fluid or leak

↓

Pump runs but can't build pressure

↓

Fluid heats up (less cooling, more friction)

↓

OVERHEAT light

↓

System degrades further

🧠 Quick diagnostic logic

Step 1: Are pumps ON?

- YES (in your panel)

Step 2: Still **LOW PRESSURE**?

- YES

👉 That means:

The system cannot pressurize

Step 3: Most likely causes

- Hydraulic leak ✓
- Low reservoir quantity ✓
- Pump damage ✓

✈️ System impact

Depending on which system (A or B):

You can lose:

- Flight control redundancy
- Landing gear operation (System A primary)
- Brakes / anti-skid (System B components)
- Nose wheel steering

Important comparison

Condition	Meaning
LOW PRESSURE + pump OFF	Normal
LOW PRESSURE + pump ON	System failure ⚠️
LOW PRESSURE + OVERHEAT	Major problem (likely fluid/pump issue) 🚨

⚡ Simple way to remember

LOW PRESSURE = “no hydraulic power being produced”

☒ Summary

- **LOW PRESSURE (amber)** = pump not producing required hydraulic pressure
- Causes:
 - Pump failure
 - Low fluid
 - Leak
- With your panel: 👉 **Combined with OVERHEAT = serious hydraulic system malfunction**

Big-picture (your scenario)

Across your panels, you now have:

-  Bleed system failure
-  Pack shutdown
-  Temperature control issues
-  Hydraulic system degradation

👉 This is essentially a **multi-system failure scenario**, and on the line:

 **LOW PRESSURE + OVERHEAT together = focus on hydraulic fluid quantity and leaks FIRST**

WINDOW OVERHEAT

WINDOW OVERHEAT Light (Amber) — What it means on the 737

Looking at your image, you have:

- **OVERHEAT lights illuminated** above both **L and R window heat sections**
- The **ON (green)** lights are also illuminated → window heat is powered

Definition

The **WINDOW OVERHEAT light** comes on when a **flight deck window has exceeded its temperature limit**.

What the window heat system does

The 737 uses **electrically heated cockpit windows** to:

- Prevent icing
- Prevent fogging
- Maintain structural integrity of the glass

Each side has:

- **Side windows**
- **Forward windows**
- Controlled automatically (A/B controllers)

What causes WINDOW OVERHEAT

1. **Overtemperature condition**

- Heating element drives temperature too high
- Could be due to:
 - Control circuit failure
 - Sensor problem

2. **Temperature controller failure**

- Auto controller (A or B) not regulating properly
- Allows window heat to exceed limit

3. **Sensor malfunction**

- False reading or no feedback
- System cannot regulate → overheats

4. **Electrical fault**

- Short or runaway heating condition

System response (very important)

When OVERHEAT is detected:

1.  The system **automatically removes power** to that window
2.  **OVERHEAT light illuminates (amber)**
3.  Protects window from damage

 The green "ON" lights you see indicate power is/was applied — but the system is protecting components

Your panel interpretation

You have:

- **Both L and R OVERHEAT lights ON**
- **Both sides ON (green)**

 That suggests:

 **System-level issue affecting both window heat controllers or inputs**

Because:

- It is unlikely that multiple windows independently overheat at the same time

Line maintenance thinking

If ONE OVERHEAT light:

👉 Local issue (window, sensor, or element)

If MULTIPLE (like your panel):

👉 Think:

- Controller problem (A/B system)
- Power/control fault
- Common input issue

Important distinction

Light	Meaning
ON (green)	Window heat powered
OVERHEAT (amber)	Temp exceeded limit, protection activated

Simple way to remember

WINDOW OVERHEAT = “window got too hot, system shut it down”

Operational effect

- Window heat is:
 - Reduced or removed on affected windows
- Risks:
 - Window icing
 - Reduced visibility
 - Structural stress (in extreme cases)

Summary

- **OVERHEAT (amber)** = window exceeded safe temp
- Causes:
 - Controller failure
 - Sensor issue

- Electrical fault
- In your image: 🖐️ **Both sides = likely system/control issue, not isolated window failures**

💡 Big-picture connection

This is a **completely separate system** from:

- Bleed air
- Packs
- Hydraulics

🖐️ It belongs to **anti-ice / electrical heating systems**

CAPT PITOT

📺 **CAPT PITOT Light (Amber) — What it means on the 737**

Looking at your image, the **CAPT PITOT light (amber)** is illuminated on the left side of the probe heat panel.

☑️ **Definition**

The **CAPT PITOT light** indicates a **problem with the captain's pitot probe heating system**.

🔑 **What the pitot system does**

The **pitot probe** measures:

- **Total air pressure**
- Used to calculate:
 - **Airspeed**

Each side has its own:

- Pitot probe
- Heating element

🖐️ Heating prevents:

- Ice buildup

- Blockage of pressure ports

What triggers the CAPT PITOT light

1. Heater failure (most common)

- Heating element not working
- Open circuit or burned-out heater

2. Electrical issue

- Loss of power to heater
- Wiring problem

3. Overheat or control problem

- Temperature regulation issue
- System detects abnormal condition (too hot or not heating properly)

4. Probe control fault

- Auto control system not functioning correctly

System logic (important)

When the light comes on:

The system has detected that the **pitot heat is not operating as expected**

 It could mean:

- Not heating when it should
- Drawing incorrect current
- Overheating or underheating

What your panel shows

You have:

- **CAPT PITOT light ON**
- Also multiple probe heat lights (L/R alpha vane, temp probe, etc.)

🔑 This suggests:

⚠️ **A broader probe heat system issue or multiple heater faults**

Because:

- It's not just one isolated probe

🧠 Line maintenance interpretation

If ONLY CAPT PITOT light:

👉 Likely:

- Captain's pitot heater failed

If MULTIPLE probe lights (like your panel):

👉 Think:

- Power supply issue
- Control system fault
- Common electrical problem

✈️ Operational impact

If captain's pitot probe is not heated:

- Risk of:
 - **Pitot icing**
 - **Blocked airspeed indication**

👉 Could lead to:

- Unreliable airspeed on captain's instruments

🔑 Important comparison

Light	Meaning
CAPT PITOT	Captain's pitot heater fault

L/R ELEV PITOT, ALPHA VANE, TEMP PROBE	Other probe heater faults
OVERHEAT (window)	Window heating issue

⚡ Simple way to remember

CAPT PITOT = “captain’s airspeed probe heat not working properly”

☑ Summary

- **CAPT PITOT (amber)** = fault in captain’s pitot heat system
- Causes:
 - Heater failure
 - Electrical issue
 - Control fault
- In your image: 🖱 Likely part of a **larger probe heat system issue**

💡 Big-picture from your panel

You now have issues across:

- Bleed air ✓
- Pressurization ✓
- Hydraulics ✓
- Window heat ✓
- Probe heat ✓

🖱 This looks like a **complex multi-system electrical or configuration scenario**, not just isolated faults.

L ELEV LIGHT

▨ L ELEV PITOT Light (Amber) — What it means on the 737

Looking at your panel, the **L ELEV PITOT light (amber)** is illuminated on the left column of the probe heat section.

☑ Definition

The **L ELEV PITOT light** indicates a **fault in the left elevator pitot probe heating system**.

What the “elevator pitot” is

The 737 has additional **pitot probes on the horizontal stabilizer (elevator area)**.

These are used primarily for:

- **Air data redundancy**
- Feeding input to:
 - Air data computers
 - Stall warning / flight control systems

 They are separate from:

- Captain’s pitot
- First Officer’s pitot

What triggers the L ELEV PITOT light

The light comes on when the system detects:

1. **Heater not working**

- Element failed (most common)
- Open circuit

2. **Electrical issue**

- Loss of power
- Wiring fault

3. **Abnormal current draw**

- Too much or too little current
- Indicates:
 - Heater malfunction
 - Possible internal damage

4. **Control system issue**

- Fault in probe heat control logic (AUTO system)

System behavior

When the light illuminates:

The system has detected that the **heater is not functioning properly**

👉 It could mean:

- Not heating (most critical)
- Or abnormal operation (overheat / current issue)

Your panel context

From your image, you also have:

- **CAPT PITOT**
- **R ELEV PITOT**
- **ALPHA VANE**
- **TEMP PROBE**

👉 Multiple probe heat lights are ON

👉 That tells you:

⚠️ **This is NOT just one probe failure**

It strongly suggests:

- **Common power or control issue**, or
- Multiple heaters affected at once

Line maintenance interpretation

If ONLY L ELEV PITOT:

👉 Likely:

- That specific probe heater failed

If MULTIPLE probe heat lights (your case):

👉 Think:

- Probe heat **system-wide problem**:

- Power supply issue
- Control circuit fault
- Possibly AUTO control problem

✈ Operational impact

If elevator pitot heaters fail:

- Those probes can:
 - Ice up
 - Provide unreliable data

👉 Effects:

- Loss of redundancy
- Possible impacts to:
 - Air data systems
 - Stall warning inputs

🔑 Important comparison

Light	Meaning
CAPT PITOT	Captain's primary pitot heat fault
L ELEV PITOT	Left elevator pitot heat fault
R ELEV PITOT	Right elevator pitot heat fault
ALPHA VANE	Angle-of-attack sensor heat fault

⚡ Simple way to remember

L ELEV PITOT = "left tail pitot probe heat not working properly"

☑ Summary

- **L ELEV PITOT (amber)** = fault in left elevator pitot heater
- Causes:
 - Heater failure
 - Electrical fault
 - Control issue
- In your panel: 👉 Likely part of a **larger probe heat system problem, not isolated**

💡 Big-picture insight

Across all your probe heat lights:

⚠️ When **multiple probe heat lights come on together**, it almost always points to:

- A **shared power source issue**
- Or **system control logic problem**, not individual probe failures

TEMP PROBE LIGHT

📺 TEMP PROBE Light (Amber) — What it means on the 737

Looking at your panel, the **TEMP PROBE light (amber)** is illuminated on the left side of the **probe heat section**.

✅ Definition

The **TEMP PROBE light** indicates a **fault in the Total Air Temperature (TAT) probe heating system**.

🔑 What the temperature probe does

The **TAT probe** measures:

- **Outside air temperature (corrected for airspeed)**

It feeds data to:

- Air Data Computers (ADC)
- FMC
- Engine and performance calculations

👉 This is **critical for accurate aircraft performance**

❄️ Why it needs heat

- The probe is exposed to airflow and moisture
- It can **ice up quickly**

- So it has an **electric heater**

What triggers the TEMP PROBE light

The light comes on when the system detects:

1. **Heater failure (most common)**

- Heating element burned out
- Probe not heating properly

2. **Electrical issue**

- Loss of power
- Wiring problem
- Relay failure

3. **Abnormal current draw**

- Too high or too low current
- Indicates heater malfunction

4. **Control system issue**

- AUTO heat logic not functioning properly

System logic

When illuminated:

The system has detected that the **TAT probe heater is not operating correctly**

 Could mean:

- Not heating → icing risk
- Or abnormal operation

Your panel (very important)

You have:

- **TEMP PROBE light ON**

- Plus:
 - **CAPT PITOT**
 - **L/R ELEV PITOT**
 - **ALPHA VANE**

👉 This tells you:

⚠️ **This is NOT a single probe failure**

It strongly indicates:

- A **common probe heat system problem**

🗣️ **Line maintenance interpretation**

If ONLY TEMP PROBE:

👉 Likely:

- TAT probe heater failed

If MULTIPLE probe lights (your case):

👉 Think:

- Shared system issue:
 - Power supply
 - Control logic (AUTO mode)
 - Electrical distribution problem

✈️ **Operational effect**

If TAT probe heat fails:

- Probe may ice up
- Results in:
 - Incorrect temperature readings

👉 This affects:

- Air data calculations
- Engine performance data
- FMC accuracy

Important comparison

Light	Meaning
TEMP PROBE	TAT probe heat fault
CAPT PITOT	Captain's pitot heat fault
ELEV PITOT	Tail pitot heat fault
ALPHA VANE	AOA sensor heat fault

Simple way to remember

TEMP PROBE = "outside temp sensor heater not working properly"

Summary

- **TEMP PROBE (amber)** = fault in TAT probe heater
- Causes:
 - Heater failure
 - Electrical issue
 - Control fault
- In your image:  Part of a **multi-probe heat system issue (not isolated)**

Big-picture insight (very important)

When you see:

- **Multiple probe heat lights all at once**

 Think FIRST:

Common power source or AUTO control problem

NOT multiple sensor failures at the same time

AUX PITOT LIGHT

▣ AUX PITOT Light (Amber) — What it means on the 737

Looking at your panel, the **AUX PITOT light (amber)** is illuminated on the **right side of the probe heat section**.

☑ Definition

The **AUX PITOT light** indicates a **fault in the auxiliary pitot probe heating system**.

🔗 What the “aux pitot” is

The **auxiliary pitot probe** is an **additional air data source** separate from:

- Captain's pitot
- First Officer's pitot
- Elevator pitot probes

👉 It provides **backup airspeed/pressure data** to certain systems.

❄ Why it needs heat

Like all pitot probes:

- It is exposed to moisture and airflow
- It can **ice up**, causing:
 - Blockage
 - Incorrect pressure readings

👉 So it uses an **electrical heater**

⚠ What triggers the AUX PITOT light

The light comes on when the system detects:

1. ✖ Heater failure (most common)

- Heating element not working
- Open circuit

2. 🔧 **Electrical problem**

- Loss of power
- Wiring or relay issue

3. 🔑 **Abnormal current draw**

- Too high or too low current
- Indicates heater malfunction

4. ⚙️ **Control issue (AUTO system)**

- Probe heat logic not commanding correctly

📄 **System logic**

When illuminated:

The system senses the **aux pitot heater is not operating within limits**

👉 Could mean:

- Not heating (icing risk)
- Or abnormal operation (overheat / electrical issue)

🔗 **Your panel context (VERY important)**

You now have multiple probe heat lights:

- **CAPT PITOT**
- **L/R ELEV PITOT**
- **ALPHA VANE**
- **TEMP PROBE**
- **AUX PITOT**

👉 This tells you:

⚠️ **This is NOT multiple individual probe failures**

Instead, it strongly indicates:

- ✅ **Common power problem**

-  **Probe heat AUTO control issue**
-  **Electrical distribution fault**

Line maintenance interpretation

If ONLY AUX PITOT light:

 Likely:

- Aux pitot heater failed

If MANY probe heat lights (your scenario):

 Think FIRST:

- Power supply (bus issue)
- Control system fault (AUTO logic)
- Shared relay or circuit problem

Operational impact

If aux pitot heat fails:

- That probe can ice up
- Effects:
 - Loss of redundancy
 - Possible unreliable backup data

 Usually less critical than captain/FO pitot, but still important

Quick comparison

Light	Meaning
CAPT PITOT	Captain's airspeed source heater fault 
F/O PITOT	First Officer's heater fault 
ELEV PITOT	Tail probe heater fault
AUX PITOT	Backup probe heater fault
TEMP PROBE	Temperature probe heater fault

⚡ Simple way to remember

AUX PITOT = “backup pitot probe heat not working properly”

☑ Summary

- **AUX PITOT (amber)** = fault in auxiliary pitot heater
- Causes:
 - Heater failure
 - Electrical issue
 - Control fault
- In YOUR panel: 🖐 Part of a **system-wide probe heat problem, not isolated failures**

💡 Big-picture insight (this is key for AMTs)

When you see:

5–6 probe heat lights ON at once

🖐 Don't chase individual probes first

✅ Go straight to:

- Power source (AC transfer buses)
- Probe heat AUTO control system
- Relays/common circuits

If you want, I can walk you through **exactly which buses power each probe and how to isolate the fault in under 2–3 steps on the line** — that's where this really clicks.

REVERSER LIGHT

REVERSER Light (Amber) — What it means on the 737

In your image, both **ENGINE 1 and ENGINE 2 REVERSER lights (amber)** are illuminated on the engine panel.

☑ Definition

The **REVERSER light (amber)** indicates that the **thrust reverser system is NOT in the commanded or safe position**.

🔑 What the thrust reverser does

On the 737:

- Thrust reversers redirect engine thrust **forward**
- Used to:
 - Slow the aircraft after landing

Each engine has its own reverser system.

⚠ When the REVERSER light comes on

1. ✖ Reverser not fully stowed (very important)

- Sleeves/buckets not fully closed
- Most critical condition

👉 This is the **primary concern in flight**

2. 🔄 Reverser is in transit

- Moving between stowed and deployed
- Light remains on until fully in position

3. ⚙ Locking system issue

- Mechanical locks not engaged properly
- Position sensors don't agree

4. 🗣 Control or indication fault

- Proximity switch disagreement
- Electrical indication issue

System logic

The light illuminates when:

The system does **not see a fully stowed and locked reverser**

It uses:

- Position sensors
- Lock status

Critical difference: Ground vs Flight

On the ground (normal situations):

- Light may be ON:
 - During deployment
 - During stow sequence

 Can be **normal while transitioning**

In flight:

If REVERSER light comes ON:

 Treat as **potential uncommanded or unsafe reverser condition**

Even though 737 has:

- **Mechanical locks**
- **Hydraulic isolation**

 Reverser deployment in flight is extremely unlikely, but still treated seriously

Your panel scenario

Both REVERSER lights are ON.

Possible interpretations:

- Both reversers:
 - Not fully stowed
 - In transition
 - Or indication/system issue

 Seeing **both at once** often points to:

- System/indication issue
- Not both reversers physically failing simultaneously

Line maintenance mindset

If ONE reverser light:

👉 Focus on that engine:

- Actuator
- Locks
- Position sensors

If BOTH lights (your case):

👉 Think:

- Common issue:
 - Indication system
 - Electrical problem
 - Logic input issue

Important comparison

Light	Meaning
REVERSER (amber)	Not stowed/locked or in transit ⚠️
ENGINE CONTROL	Engine control system issue
LOW PRESSURE (hyd)	Hydraulic pressure problem

Simple way to remember

REVERSER = “not fully stowed and locked”

Summary

- **REVERSER (amber)** = thrust reverser not in safe/stowed position
- Causes:
 - Not stowed
 - In transit

- Faulty indication or locking
- In your image: 🖐 Both engines = likely **indication or system issue vs both reversers failing physically**

💡 Big-picture tie-in

You've now looked at:

- Bleed system ✓
- Hydraulics ✓
- Probe heat ✓
- Fire detection ✓
- Reversers ✓

🖐 You're essentially building a **full aircraft systems mental map**, which is exactly how real troubleshooting works:

Look at patterns — not just individual lights

ENGINE CONTROL

📺 ENGINE CONTROL Light (Amber) — What it means on the 737

In your image, both **ENGINE 1 and ENGINE 2 ENGINE CONTROL lights (amber)** are illuminated next to the EEC switches.

☑ Definition

The **ENGINE CONTROL light** indicates a **fault in the Electronic Engine Control (EEC) system OR that the engine is no longer being fully controlled electronically.**

🔑 What the EEC does

The **EEC (Electronic Engine Control)** is basically the **"computer" that controls the engine:**

- Controls fuel flow

- Regulates thrust
- Protects against limits (overtemp, overspeed)
- Optimizes engine performance

⚠️ What triggers the ENGINE CONTROL light

1. ✖ Loss of normal EEC control (most important)

- The EEC is no longer controlling the engine in its **normal mode**
- Engine reverts to a **backup (alternate) control mode**

2. ⚡ Electrical power issue

- EEC loses its normal power source
- Switches to alternate power or degraded operation

3. ⚙ Internal EEC fault

- Computer detects:
 - Invalid data
 - Internal malfunction

4. 💡 EEC in ALTN mode

Looking at your image:

- The EEC switches show **ALTN**

👉 That means:

The engines are being controlled in **alternate mode**, not normal

📖 What “ALTN mode” means

In ALTN mode:

- Engine still runs ✅
- But:
 - Less precise control
 - Reduced protections
 - Different thrust response

👉 Basically:

Engine is operating, but not fully optimized or protected

🔧 Your panel interpretation (VERY important)

You have:

- **ENGINE CONTROL lights ON (both engines)**
- **EEC switches showing ALTN**

👉 That tells you:

⚠️ **Both engines have lost normal EEC control and are operating in alternate mode**

🗣️ Line maintenance mindset

If ONE ENGINE CONTROL light:

👉 Focus on:

- That engine's EEC
- Power source
- Wiring

If BOTH lights (your case):

👉 Strongly suggests:

- ✅ **Common electrical issue**
- ✅ **Power supply problem to both EECs**
- ✅ Not likely both engines failed independently

✈️ Operational impact

In ALTN mode:

- Engine still produces thrust ✅
- But:
 - Fuel scheduling less refined

- Possible higher fuel burn
- Reduced protection margins

👉 Aircraft is still flyable, but with degraded engine control

🔑 Important comparison

Indication	Meaning
ENGINE CONTROL (amber)	EEC not in normal mode ⚠️
EEC NORMAL (no light)	Full electronic control ✅
EEC ALTN (switch)	Backup control mode active

⚡ Simple way to remember

ENGINE CONTROL = “engine computer not in normal control mode”

☑ Summary

- **ENGINE CONTROL (amber)** = EEC fault or alternate mode operation
- Causes:
 - Power loss
 - EEC fault
 - Electrical issue
- In YOUR image: 👉 **Both engines in ALTN mode → likely common electrical/system issue**

💡 Big-picture pattern (this is key)

Across ALL your panels you’ve shown:

- Probe heat failures ✅
- Window heat overheat ✅
- Engine control in ALTN ✅
- Multiple systems affected ✅

👉 This points to:

⚠️ **A common electrical power or control system problem affecting multiple aircraft systems**

NOT:

- Independent component failures everywhere

PASS OXY ON

PASS OXY ON Light (Amber) — What it means on the 737

Looking at your image, the **PASS OXY ON light (amber)** is illuminated next to the **PASS OXYGEN switch**.

Definition

The **PASS OXY ON light** indicates that the **passenger oxygen system has been activated and the masks are deployed**.

What the passenger oxygen system does

The system provides **oxygen to passengers and cabin crew** in case of:

- Cabin depressurization
- High cabin altitude

When activated:

- Oxygen masks **drop from the overhead panels**
- Chemical oxygen generators start supplying oxygen

What triggers PASS OXY ON

1. **Automatic deployment (most critical)**

- Occurs when:
 - **Cabin altitude reaches ~14,000 ft**

 System automatically:

- Releases masks
- Turns ON this light

2. 🖐️ **Manual activation (crew command)**

- Crew moves the switch from:
 - **NORMAL → ON**

👉 This:

- Forces mask deployment
- Illuminates **PASS OXY ON**

📋 **Important system behavior**

Once triggered:

- ✅ Oxygen generators **ignite chemically**
- ✅ Oxygen flows to masks
- ❗ Cannot be turned off or reset in flight

💡 **Line maintenance insight (very important)**

If you see **PASS OXY ON** in a write-up:

You must assume:

⚠️ **Passenger oxygen masks have deployed**

What that means for maintenance:

- Oxygen generators may be:
 - **Used (expended)**
- System requires:
 - **Inspection**
 - **Generator replacement**
 - Mask reset/repack

🔗 **Your panel context**

Earlier, your scenario included:

- Pressurization issues
- Bleed/pack failures

🔗 That ties directly:

Bleed/pack failure



Loss of pressurization



Cabin altitude increases



~14,000 ft reached



Passenger oxygen deploys



PASS OXY ON light

🔑 Important distinction

Light	Meaning
PASS OXY ON	Masks deployed ✅
CABIN ALT warning horn	Cabin altitude too high 🚨
AUTO FAIL (pressurization)	Controller problem

⚡ Simple way to remember

PASS OXY ON = “masks are down and providing oxygen”

☑ Summary

- **PASS OXY ON (amber)** = passenger oxygen system activated
- Causes:
 - Automatic high cabin altitude (~14,000 ft)
 - Manual crew activation
- Effects:
 - Masks deployed
 - Oxygen flowing
 - System must be serviced after use

💡 Big-picture (your full scenario)

Putting everything together from your panels:

⚠️ **Bleed/pressurization failure → cabin altitude rises → oxygen system activates → PASS OXY ON**

GPS LIGHT

📺 GPS Light (Amber) — What it means on the 737

Looking at your image, the **GPS light (amber)** is illuminated on the aft overhead panel, near the IRS controls.

✅ Definition

The **GPS light** indicates that the **GPS position input to the navigation system is unreliable or has been lost**.

🔑 What GPS does on the 737

The GPS feeds position data to:

- **FMC (Flight Management Computer)**
- **IRS (Inertial Reference System) updating**
- Navigation displays (map position accuracy)

👉 It helps keep the aircraft's position **highly accurate**

⚠️ What triggers the GPS light

1. 📶 Loss of GPS signal

- Satellites not available
- Antenna blockage or failure

2. ✖️ GPS receiver failure

- Internal fault in GPS unit

3. **Low accuracy / integrity**

- Signal quality too poor to trust
- System rejects GPS data

4. **Power or interface issue**

- Data not reaching FMC/IRS properly

System behavior

When the light illuminates:

The aircraft is **no longer using GPS as a reliable navigation source**

 What happens next:

- FMC reverts to:
 - **IRS-only navigation**
 - Possibly radio-based updating (VOR/DME)

 Aircraft is still navigable

 But accuracy is reduced over time

Your panel context

You also have IRS-related lights showing:

- **ON DC**
- **FAULT**
- **DC FAIL**

 Combined with GPS light:

 This suggests a **navigation system problem affecting multiple inputs**

- IRS degraded 
- GPS unavailable 

 That's a **bigger nav system issue**, not just GPS alone

Line maintenance mindset

If ONLY GPS light:

👉 Likely:

- GPS antenna or receiver fault

If GPS + IRS lights (your case):

👉 Think:

- Common electrical issue
- IRS power/control problem
- Data bus/interface failure

Operational impact

Without GPS:

- Navigation relies on:
 - IRS (drifts over time)
- Effects:
 - Reduced positional accuracy
 - FMC performance degraded

👉 Long flights → errors grow without updates

Important comparison

Light	Meaning
GPS (amber)	GPS data unreliable/unavailable
IRS FAULT	Inertial system problem
ON DC / DC FAIL	IRS power supply issue

Simple way to remember

GPS light = “navigation satellite data not usable”

☑ Summary

- **GPS (amber)** = GPS signal/data unavailable or unreliable
- Causes:
 - Signal loss
 - Receiver fault
 - Power/interface issue
- In YOUR panel: 🖐️ Combined with IRS faults → **navigation system degradation, not just GPS alone**

💡 Big-picture insight (very important)

Across your panels, you now have failures in:

- Probe heat ✅
- Window heat ✅
- Engine control (EEC) ✅
- IRS ✅
- GPS ✅

🔗 That pattern strongly points to:

⚠️ **A common electrical power distribution problem (likely bus-related), not isolated component failures**

PSEU LIGHT

▨ PSEU Light (Amber) — What it means on the 737

Looking at your image, the **PSEU light (amber)** is illuminated on the **lower overhead panel**.

☑ Definition

The **PSEU light** indicates a **fault in the Proximity Switch Electronic Unit (PSEU)** or one of the systems it monitors.

What the PSEU does

The **PSEU (Proximity Switch Electronic Unit)** is basically a **central logic processor** that uses proximity switches to monitor and control several aircraft systems.

It deals with things like:

- Landing gear position (up / down / in transit)
- Gear doors
- Thrust reverser interlocks
- Ground/air sensing
- Flight control interlocks (for certain protections)

What triggers the PSEU light

The light comes on when:

The PSEU detects **invalid, conflicting, or failed inputs/outputs**

Common causes:

1. PSEU internal fault

- Unit failure
- Logic problem inside the computer

2. Proximity switch issue

- Bad or misadjusted switch
- Incorrect position signal (gear, doors, etc.)

3. Electrical / power problem

- Loss of power to PSEU
- Faulty wiring or signals

4. Disagreement between sensors

Example:

- Gear shows "down" on one sensor

- Another sensor disagrees

👉 PSEU detects conflict → light ON

🔗 Tie it to your panel (VERY important)

In your image you also have:

- **REVERSER lights**
- **ENGINE CONTROL lights**
- Other system faults from earlier panels

👉 Key insight:

⚠️ The PSEU interfaces with **landing gear and reverser logic**

So if you see:

- **REVERSER lights ON**
- **PSEU light ON**

👉 Possible connection:

- PSEU not confirming proper reverser or gear logic
- Sensor disagreement or logic failure

🧠 Line maintenance mindset

If PSEU light only:

👉 Check:

- PSEU BITE
- Proximity switches (gear, doors)

If PSEU + other lights (your scenario):

👉 Think:

- **Common system issue**
- Electrical power/input problem
- Multiple sensor inputs affected

✈ Operational considerations

- Aircraft may:
 - Lose some **ground/air logic functions**
 - Have degraded **gear or reverser interlocks**

👉 Systems may still function, but **protections may be reduced**

🔑 Important distinction

Light	Meaning
PSEU	Logic/sensor system fault ⚠
REVERSER	Reverser not stowed or locked
GEAR lights	Gear position status

⚡ Simple way to remember

PSEU = “the switch logic brain sees something wrong”

☑ Summary

- **PSEU (amber)** = fault in:
 - Proximity switch system
 - Or PSEU computer itself
- Causes:
 - Sensor disagreement
 - Electrical issue
 - Internal failure
- In YOUR scenario: 👉 Likely tied into **broader electrical/system problem affecting multiple indications**

💡 Big-picture insight (this is key)

Now combining ALL your lights across panels:

- Probe heat failures ✅
- Window heat issues ✅
- Engine control ALTN ✅
- IRS/GPS faults ✅

- Reverser warnings 
- PSEU 

 This pattern strongly points to:

 **A common electrical power or distribution failure (likely bus-related), not individual component failures**

ON DC LIGHT

 **ON DC Light (Amber) — What it means on the 737 (IRS Panel)**

Looking at your image, the **ON DC lights (amber)** are illuminated for both:

- **Left IRS**
- **Right IRS**

Definition

The **ON DC light** indicates that the **Inertial Reference System (IRS)** is being powered by **DC electrical power instead of normal AC power**.

Normal IRS power source

- The IRS normally runs on:  **AC power (from the aircraft generators)**

What ON DC means

When the **ON DC light illuminates**:

The IRS **lost its normal AC power source** and has automatically switched to **DC backup power**

Why that matters

DC power can:

-  Keep IRS memory and basic operation alive
-  **BUT cannot support full navigation capability for long**

Common causes of ON DC

1. **Loss of AC power (most common)**

- Generator problem
- Bus failure
- Transfer bus issue

2. **Electrical system fault**

- AC bus not supplying power to IRS
- Power distribution issue

3. **Power transfer event**

- Temporary switching between sources
- May be momentary

System behavior

When ON DC is active:

- IRS is:
 -  Powered
 -  **Degraded in capability**

After a short time (~5 minutes typical logic importance):

- IRS may:
 - Lose alignment
 - Lose accurate navigation capability

Your panel context (VERY important)

You also have:

- **DC FAIL lights**
- **FAULT lights**
- **GPS light**
- **Multiple system failures across aircraft**

👉 This is the key takeaway:

⚠️ **This is NOT just an IRS issue — it is a major electrical power problem**

🧠 Line maintenance reasoning

If ON DC only:

👉 Temporary power switch or AC interruption

If ON DC + DC FAIL + FAULT (your case):

👉 Strongly indicates:

- ✅ Loss of normal AC power
- ✅ IRS degraded and failing
- ✅ Electrical bus issue

🔗 Tie to **EVERYTHING** you've shown (this is the big picture)

Across your panels:

- Probe heat failures ✅
- Window heat issues ✅
- Engine control in ALTN ✅
- GPS failure ✅
- IRS on DC ✅
- PSEU faults ✅

👉 All of these systems rely on AC electrical power

So when you see:

✅ **ON DC (IRS)**

- ✅ multiple unrelated systems failing

✦ **Conclusion (this is the root cause pattern)**

⚠ **Loss of AC electrical power (likely a transfer bus or generator failure)**

🔑 **Simple way to remember**

ON DC = "IRS lost normal AC power and is running on backup DC"

☑ **Summary**

- **ON DC (amber)** = IRS powered by DC instead of AC
- Causes:
 - AC power loss
 - Electrical bus failure
- Effect:
 - Temporary operation only
 - Navigation degradation
- In YOUR scenario: 👉 **Confirms a larger electrical system failure affecting multiple aircraft systems**

💡 **Final AMT-level takeaway**

If you walk up to an aircraft and see:

- **ON DC lights + multiple unrelated system warnings**

👉 Don't chase individual components.

✅ Go straight to:

- **Electrical power distribution**
- **Generators**
- **Transfer buses**

DC FAIL LIGHT

▤ **DC FAIL Light (Amber) — What it means on the 737 (IRS Panel)**

Looking closely at your image, both **IRS panels (left and right)** show:

- ✅ **ON DC (amber)**
- ✅ **DC FAIL (amber)**

-  **FAULT (amber)**

Definition

The **DC FAIL light** indicates that the **Inertial Reference System (IRS)** is no longer receiving **DC backup power**.

How IRS power works (simple)

The IRS normally uses:

1.  **Primary power** → **AC (normal operation)**
2.  **Backup power** → **DC (temporary sustain power)**

Power failure sequence (VERY important)

Step 1 — Normal:

- IRS powered by **AC**
- Everything works normally

Step 2 — AC lost:

- **ON DC light comes ON**  IRS switches to DC backup

Step 3 — DC also lost:

- **DC FAIL light comes ON** 

 Now:

IRS is losing TOTAL electrical support

What DC FAIL really means

The IRS **cannot be sustained anymore** — it is effectively shutting down or has already failed

⚠️ What causes DC FAIL

1. 🔌 Complete power loss to IRS

- AC power gone
- DC power unavailable

2. ⚡ Electrical distribution failure

- Loss of DC bus
- Transfer logic failure

3. 🔋 Battery/backup supply issue

- DC source not feeding IRS

🔧 Your panel (CRITICAL interpretation)

You have:

- ✅ ON DC
- ✅ DC FAIL
- ✅ FAULT
- ✅ GPS light
- ✅ Multiple aircraft system failures

👉 This is the key conclusion:

⚠️ This is a TOTAL loss of usable IRS power

The system went:

AC power lost

↓

Switched to DC (ON DC)

↓

DC also lost

↓

DC FAIL

↓

IRS failure (FAULT)

Line maintenance mindset

If ON DC only:

👉 AC power problem

If ON DC + DC FAIL (your case):

👉 🚨 **Complete electrical supply failure to IRS**

If both sides affected:

👉 Not individual IRS issues

👉 ✅ **Common electrical system problem**

Tie into ALL your previous lights (this is the BIG picture)

Across everything you've shown:

- Probe heat failures ✅
- Window heat issues ✅
- Engine control ALTN ✅
- GPS failure ✅
- PSEU light ✅
- Reverser lights ✅
- IRS ON DC ✅
- IRS DC FAIL ✅

✨ **This tells you ONE thing:**

⚠️ **Major aircraft electrical failure — likely AC transfer bus or generator problem**

Operational effect

With DC FAIL:

- IRS:
 - ✗ Loses alignment
 - ✗ Stops providing navigation data
- Navigation:
 - Degraded or lost

🔑 Simple way to remember

ON DC = “lost AC, running on backup”

DC FAIL = “even backup power is gone”

☑ Summary

- **DC FAIL (amber)** = IRS lost DC backup power
- Occurs after:
 - AC power loss (ON DC first)
- Indicates:
 - **Complete IRS power failure**
- In your scenario: 👉 Confirms a **major electrical system failure affecting the whole aircraft**

💡 Final AMT-level takeaway

When you see:

- **ON DC + DC FAIL + multiple unrelated system lights**

👉 Don't troubleshoot individual systems first.

✅ Go straight to:

- Generators
- AC transfer buses
- DC buses
- Electrical distribution system

FAULT LIGHT

FAULT Light (Amber) — What it means on the 737 (IRS Panel)

Looking at your image, both **L and R IRS panels** show the **FAULT lights (amber)** illuminated.

Definition

The **FAULT light** indicates that the **Inertial Reference System (IRS)** has detected an **internal malfunction OR can no longer provide valid navigation/attitude data**.

What the IRS does (quick reminder)

The IRS provides:

- **Aircraft position (navigation)**
- **Attitude (pitch/roll)**
- **Heading and inertial data**

 It is a **core navigation system**

What triggers the FAULT light

1. **Internal IRS failure**

- Hardware fault inside the unit
- Gyros or accelerometers not functioning

2. **Loss of valid data**

- System cannot compute reliable position or attitude

3. **Power disruption (very important in your case)**

- Loss of AC power
- Loss of DC backup

 This is the **big trigger in your panel**

4. **Alignment failure**

- IRS fails to align correctly

- Incorrect or interrupted alignment process

System logic (what happened step-by-step)

From your panel, you have:

-  **ON DC**
-  **DC FAIL**
-  **FAULT**

 The sequence is:

Normal operation (AC power)

↓

Loss of AC → ON DC light

↓

Loss of DC → DC FAIL light

↓

IRS loses all power/support

↓

FAULT light illuminates

What FAULT means in THIS scenario

 The FAULT light is NOT just an IRS issue

 It is the **result of total power loss and system collapse**

Your panel (CRITICAL interpretation)

Both IRS units show:

- FAULT
- ON DC
- DC FAIL

 That means:

 **Both IRS units have effectively failed due to loss of electrical power**

Line maintenance mindset

If FAULT alone:

 Check:

- IRS unit
- Alignment
- Internal failure

If FAULT + ON DC + DC FAIL (your case):

 Do NOT chase the IRS first

 Go to:

- Electrical system
- AC power sources
- DC power sources
- Bus distribution

Tie this into ALL your previous lights (this is the big picture)

You've shown:

- Probe heat failures 
- Window heat faults 
- Engine control ALTN 
- GPS light 
- PSEU light 
- Reverser lights 
- IRS ON DC 
- IRS DC FAIL 
- IRS FAULT 

Final conclusion (this is the key skill)

 **The FAULT light is a downstream symptom — not the root cause**

The root cause is:

 **Major electrical power failure (likely AC transfer bus or generator loss)**

✈ Operational impact

With IRS FAULT:

- ❌ No inertial navigation
- ❌ No attitude reference (from IRS)
- ❌ FMC navigation degraded or lost

🔑 Simple way to remember

FAULT = "IRS has failed or cannot provide usable data"

In your case:

FAULT = "IRS died because power was lost"

☑ Summary

- **FAULT (amber)** = IRS malfunction or failure
- Causes:
 - Internal failure
 - Power loss
 - Alignment failure
- In YOUR panel: 🖐 **Caused by complete loss of AC and DC power to IRS**

💡 Final AMT takeaway

When you see:

- **ON DC → DC FAIL → FAULT**

🖐 That's a **textbook electrical failure progression**

✅ Always troubleshoot:

1. Generators
2. AC transfer buses
3. DC buses / TR units
4. Power distribution

ELT LIGHT

ELT Light (Amber) — What it means on the 737

Looking at your image, the **ELT light (amber)** is illuminated next to the **ELT switch (ARM / ON)**.

Definition

The **ELT light** indicates that the **Emergency Locator Transmitter (ELT)** has been activated or there is a fault in the ELT system.

What the ELT does

The **ELT (Emergency Locator Transmitter)** is a self-contained device designed to:

- Transmit a **distress signal** after:
 - A crash
 - A severe impact
- Helps search and rescue locate the aircraft

What triggers the ELT light

1. **ELT activated (most important)**

- Triggered by:
 - **Impact sensor (G-switch) inside the ELT**
- Automatically turns ON after a crash or hard impact

 Light comes on to indicate:

The ELT is transmitting

2. **Manual activation**

- ELT switch moved from:
 - **ARM → ON**

 Forces transmission → light comes on

3. **ELT system fault**

- Internal failure in the ELT
- Wiring or antenna issue

4. **Battery issue**

- ELT battery low or failed
- Fault detected → light may illuminate

Switch positions (important)

You have:

- **ARM (normal position)**
- **ON (manual activation)**

In NORMAL operation:

- Switch stays in **ARM**
- ELT activates **automatically only if triggered**

Your panel context

The light is ON while the switch appears in **ARM**.

 That suggests:

 **ELT has activated automatically OR there is a fault**

Line maintenance mindset

If ELT light ON on the ground:

Check:

1.  Has ELT been activated?
 - a. Listen for beacon signal
 - b. Check ELT panel or unit
2.  Possible causes:
 - a. Hard landing / shock triggered it

- b. Maintenance handling triggered sensor
- c. Internal ELT fault

✈ Operational significance

- If ELT is actually transmitting:
 - It sends a **distress signal to satellites**
- Needs to be:
 - Verified
 - Reset if accidental

🔑 Important distinction

Indication	Meaning
ELT light ON + switch ON	Manually activated
ELT light ON + switch ARM	Auto activation or fault ⚠
No light	Normal standby

⚡ Simple way to remember

ELT light = “emergency beacon is active or has a problem”

☑ Summary

- **ELT (amber)** = ELT activated or fault present
- Causes:
 - Impact activation ✓
 - Manual ON ✓
 - Fault or battery issue ✓
- In your image: 👉 Likely **auto activation or system fault** since switch is in ARM

💡 Big-picture insight

Unlike many other lights you showed:

👉 **ELT is NOT tied to aircraft electrical system failures**

(It has its own independent power source)

So:

Even if the aircraft loses power, **ELT can still activate and transmit**

LOW PRESSURE

LOW PRESSURE Light (Amber) — What you're seeing on THIS panel

Looking at your image, you actually have **two different LOW PRESSURE indications**:

1.  **System A LOW PRESSURE**
2.  **System B LOW PRESSURE**
3.  **Standby Hydraulic LOW PRESSURE (in the STBY HYD section)**

Definition (Hydraulic LOW PRESSURE)

A **LOW PRESSURE light** means that the **hydraulic system or pump is NOT producing normal pressure (~3000 psi)**.

Breaking down YOUR specific panel

1. System A & B LOW PRESSURE

Located under the flight control section:

Indicates **main hydraulic Systems A and B are not pressurized**

2. STANDBY HYD LOW PRESSURE

Located in the standby hydraulic panel:

Indicates the **standby hydraulic system is also not producing pressure**

BIG PICTURE FROM YOUR IMAGE

You have:

-  **A LOW PRESSURE**

-  **B LOW PRESSURE**
-  **STBY LOW PRESSURE**
-  **LOW QUANTITY (standby)**
-  Multiple flight control warnings:
 - FEEL DIFF PRESS
 - SPEED TRIM FAIL
 - MACH TRIM FAIL
 - AUTO SLAT FAIL
 - YAW DAMPER light

What this means (CRITICAL)

 **ALL hydraulic systems are essentially lost or severely degraded**

This is **not normal at all** and indicates:

 **Complete or near-complete hydraulic failure**

What causes LOW PRESSURE (per system)

1. **Pump not running**

- Engine-driven pump not turning
- Electric pump off or failed

2. **Low fluid quantity**

- You have:
 - **STBY LOW QUANTITY**

 Strong sign of:

Hydraulic fluid loss (possible leak)

3. **System leak**

- Fluid escaping → pressure lost
- Affects:
 - A
 - B

- Standby if severe

4. **Mechanical failure**

- Pump failure
- Drive failure

Line maintenance interpretation

Single system LOW PRESSURE:

 Isolated issue

Multiple systems (your panel):

 Think:

- **Major leak** 
- **Power loss to pumps** 
- **System-wide failure** 

Tie to other lights on THIS panel

You also have:

- **STBY RUD ON**
- **FEEL DIFF PRESS**
- **SPEED TRIM FAIL**
- **MACH TRIM FAIL**
- **AUTO SLAT FAIL**
- **YAW DAMPER**

 That's because hydraulics power:

- Flight controls
- Rudder
- Trim systems
- Slats
- Dampers

So this chain happens:

Hydraulic pressure lost



Flight control systems lose power



Backup / standby activates



Multiple control warnings appear

✈ Operational impact

With this condition:

- ❌ Normal flight controls degraded
- ❌ Autopilot likely affected
- ❌ Slats/flaps function limited
- ✅ Aircraft still controllable (with backup), but **seriously degraded**

🔑 Important takeaway

Condition	Meaning
ONE LOW PRESSURE	Local problem
MULTIPLE LOW PRESSURE	System-wide issue ⚠
ALL systems LOW	Major failure 🚨

⚡ Simple way to remember

LOW PRESSURE = “no hydraulic power”

ALL LOW = “no airplane systems working normally”

☑ Summary (your exact panel)

- **System A LOW PRESSURE** → no A pressure
- **System B LOW PRESSURE** → no B pressure
- **Standby LOW PRESSURE** → backup also failing

- **LOW QUANTITY (standby)** → likely fluid loss

💡 Final AMT-level insight

This panel is showing:

⚠️ **Not just a pump issue — likely fluid loss or major system failure affecting ALL hydraulics**

YAW DAMPER LIGHT

▣ YAW DAMPER Light (Amber) — What it means on the 737

Looking at your image, the **YAW DAMPER light (amber)** is illuminated on the **flight control panel**, and the **yaw damper switch appears OFF**.

☑ Definition

The **YAW DAMPER light** indicates that the **yaw damper system is not operating** or has **failed / been turned off**.

🔑 What the yaw damper does

The **yaw damper** automatically controls the **rudder** to:

- Prevent **Dutch roll oscillations**
- Smooth out yaw (side-to-side motion)
- Improve **stability and passenger comfort**

👉 It uses:

- Yaw rate sensors
- Flight control computers
- Hydraulic rudder input

⚠️ Why the YAW DAMPER light comes on

1. ☑ Switch OFF (what your image shows)

- The yaw damper switch is in **OFF**

- Light illuminates to confirm:

“Yaw damper is not active”

👉 This is **normal when OFF is selected**

2. ❌ **System failure**

- Yaw damper fails internally
- Computer issue or servo problem

3. ⚡ **Electrical power issue**

- Loss of power to yaw damper system

4. 💧 **Hydraulic problem (very relevant to your panel)**

- Yaw damper uses **hydraulic System B**
- If System B has:
 - **LOW PRESSURE** (which you DO have)

👉 Then:

Yaw damper cannot operate → light ON

🔗 **Tie it to YOUR panel (VERY important)**

You also have:

- ✅ **Hydraulic LOW PRESSURE (A & B)**
- ✅ **Standby hydraulic issues**
- ✅ Multiple flight control warnings

👉 So in your case:

⚠️ The yaw damper is OFF / not working **because hydraulic pressure is lost**

📋 **Failure chain (what's happening)**

Hydraulic pressure lost (System B)

↓

Rudder control degraded
↓
Yaw damper cannot function
↓
YAW DAMPER light ON

✈ Operational impact

Without yaw damper:

- Aircraft may experience:
 - **Dutch roll (oscillations)**
 - Increased pilot workload
 - Less stability

👉 Aircraft is still controllable ✅

But:

Handling becomes more demanding

🔑 Important distinction

Condition	Meaning
Light ON + switch OFF	Normal (system intentionally off)
Light ON + switch ON	System failure ⚠
Light ON + hyd issues	Loss of hydraulic power ⚠

⚡ Simple way to remember

YAW DAMPER light = “rudder stabilization system not working”

☑ Summary (your exact case)

- **YAW DAMPER light ON**
- Switch is **OFF**
- Hydraulic systems show **LOW PRESSURE**

👉 Conclusion:

- ✅ The yaw damper is OFF
- ✅ And likely cannot operate anyway due to **hydraulic failure**

💡 Big-picture insight

Your panel is showing:

⚠️ **A major hydraulic system failure cascading into flight control system degradation**

And the yaw damper light is: 👉 **One symptom of that larger failure**

FEEL DIFF PRESS

⚠️ What triggers FEEL DIFF PRESS

The light comes on when:

❌ The system does **not detect enough pressure difference between Hydraulic System A and B**

🔍 Common causes

1. 💧 Loss of one or both hydraulic systems (MOST relevant to your panel)

- You have:
 - ✅ **A LOW PRESSURE**
 - ✅ **B LOW PRESSURE**

👉 That means:

No differential pressure is possible

2. ⚙️ Feel unit malfunction

- Internal failure in feel/centering unit

3. 📡 Sensor or indication issue

- Pressure sensing disagreement

🔧 Tie it to YOUR panel (CRITICAL)

From your image:

- ✅ **A LOW PRESSURE**
- ✅ **B LOW PRESSURE**
- ✅ **Standby hydraulic issues**
- ✅ Multiple flight control warnings

👉 So what's happening:

Hydraulic systems A & B lost

↓

No pressure to feel system

↓

No differential pressure available

↓

FEEL DIFF PRESS light ON

✈️ Operational effect

With FEEL DIFF PRESS:

- ❌ Pilot loses normal **control feel**
- Control column may feel:
 - Too light
 - Inconsistent

👉 In extreme cases:

- Risk of **overcontrolling the aircraft**

🧠 Line maintenance interpretation

If FEEL DIFF PRESS alone:

👉 Check:

- Hydraulic supply
- Feel unit

If FEEL DIFF PRESS + HYD LOW PRESSURE (your case):

👉 Don't chase the feel system

✅ Root cause is:

Loss of hydraulic pressure

💡 Important takeaway

Condition	Meaning
FEEL DIFF PRESS only	Feel system issue
FEEL DIFF PRESS + HYD issues	Hydraulic failure causing it ✅

⚡ Simple way to remember

FEEL DIFF PRESS = "no hydraulic pressure difference to simulate control feel"

✅ Summary (your exact scenario)

- **FEEL DIFF PRESS (amber)** = insufficient differential pressure
- Caused by:
 - ✅ Loss of Hyd System A
 - ✅ Loss of Hyd System B
- Result:
 - Degraded or abnormal control feel

💡 Big-picture (this panel as a whole)

Your entire flight control section shows:

⚠️ **Full hydraulic system failure** → **cascading loss of flight control support systems**

And the FEEL DIFF PRESS light is: 👉 **A direct symptom of that hydraulic failure**

SPEED TRIM FAIL

SPEED TRIM FAIL Light (Amber) — What it means on the 737

Looking at your panel, the **SPEED TRIM FAIL light (amber)** is illuminated in the **flight control warning section**, along with:

- **MACH TRIM FAIL**
- **AUTO SLAT FAIL**
- **FEEL DIFF PRESS**
- **HYD LOW PRESSURE**

Definition

The **SPEED TRIM FAIL light** indicates that the **Speed Trim System is not operating properly or has been disabled**.

What the Speed Trim System does

The **Speed Trim System (STS)** automatically moves the **stabilizer** to:

- Improve **longitudinal stability**
- Assist pilot control at:
 - **Low speeds**
 - **High thrust**
 - **Low weight / aft CG conditions**

 It prevents:

- Pitch instability
- Nose-up tendencies

When SPEED TRIM is active

Normally operates:

- Flaps UP
- Low airspeed
- High thrust
- Autopilot OFF

⚠️ What causes SPEED TRIM FAIL

1. 💧 Hydraulic pressure loss (MOST IMPORTANT in your case)

Speed trim uses:

- **Stabilizer trim system**
- Requires hydraulic power (primarily System B components)

👉 You have:

- ☒ **Hydraulic A LOW PRESSURE**
- ☒ **Hydraulic B LOW PRESSURE**
- ☒ **Standby low pressure**

2. ⚙️ Stabilizer trim system failure

- Trim motor or actuator fault

3. ⚡ Electrical/control failure

- Fault in speed trim computer logic

4. ❌ Invalid data inputs

- Airspeed, thrust, or sensor disagreement

🔗 Your panel (CRITICAL interpretation)

You have ALL of these together:

- **LOW PRESSURE (A & B)**
- **STBY LOW PRESSURE**
- **FEEL DIFF PRESS**
- **SPEED TRIM FAIL**
- **MACH TRIM FAIL**
- **AUTO SLAT FAIL**
- **YAW DAMPER**

👉 That tells you:

⚠️ **Speed Trim did NOT fail on its own — it was LOST because hydraulics failed**

Failure chain (what's happening)

Hydraulic pressure lost (A, B, Standby)



Stabilizer & flight control functions degrade



Speed Trim system cannot operate



SPEED TRIM FAIL light ON

Operational effect

With SPEED TRIM FAIL:

-  No automatic stabilizer trim assistance
- Aircraft may:
 - Be harder to control at low speed
 - Require more manual trim input

 Aircraft is still controllable 

But:

Pilot workload increases significantly

Line maintenance mindset

If SPEED TRIM FAIL alone:

 Check:

- Trim system
- STS computer
- Electrical inputs

If SPEED TRIM FAIL + HYD issues (your panel):

 Root cause is:

 **Loss of hydraulic power**

Do NOT troubleshoot STS first

🔑 Important connection (with other lights)

Light	Why it's also on
MACH TRIM FAIL	Same stabilizer dependency
AUTO SLAT FAIL	Hydraulic/slat system failure
FEEL DIFF PRESS	No hydraulic differential pressure
YAW DAMPER	Needs hydraulic pressure

⚡ Simple way to remember

SPEED TRIM FAIL = “automatic pitch stability assist is gone”

☑ Summary (your exact scenario)

- **SPEED TRIM FAIL (amber)** = speed trim system not working
- Root cause in your case:
 - ☒ Loss of Hydraulic System A
 - ☒ Loss of Hydraulic System B
 - ☒ Standby system degraded

👉 It is a **result of a major hydraulic failure**, not a standalone problem

💡 Big-picture insight

Your panel represents a **classic cascade failure**:

⚠ **Hydraulic failure** → **flight control systems degrade** → **multiple “FAIL” lights appear**

And:

👉 **SPEED TRIM FAIL is just one symptom of that larger failure**

MACH TRIM FAIL

📊 MACH TRIM FAIL Light (Amber) — What it means on the 737

Looking at your panel, the **MACH TRIM FAIL light (amber)** is illuminated along with:

- **SPEED TRIM FAIL**
- **AUTO SLAT FAIL**
- **FEEL DIFF PRESS**
- **HYD LOW PRESSURE (A & B)**

Definition

The **MACH TRIM FAIL light** indicates that the **Mach Trim System is inoperative or unable to function**.

What the Mach Trim System does

The **Mach Trim System (MTS)** automatically adjusts the **horizontal stabilizer** to:

- Compensate for **Mach tuck** (nose-down tendency at high speed)
- Maintain **longitudinal stability at high Mach numbers**

 It works mainly:

- At **high altitude**
- At **high airspeeds (high Mach)**

Why Mach Trim is important

At high speed:

- Aircraft tendency:
 - Nose pitches **down** (Mach tuck)

 Mach Trim:

- Automatically adds **nose-up trim**
- Keeps the aircraft stable and controllable

What triggers MACH TRIM FAIL

1.  **Hydraulic pressure loss (MOST important in your scenario)**

Mach Trim uses:

- **Stabilizer trim system**
- Requires **hydraulic power** (primarily System B related components)

👉 In your panel:

-  **Hyd A LOW PRESSURE**
-  **Hyd B LOW PRESSURE**
-  **Standby issues**

2. **Stabilizer trim failure**

- Trim actuator or motor failure

3. **Electrical/control failure**

- Mach trim computer failure
- Loss of input signals (air data)

4. **Invalid air data inputs**

- Incorrect Mach number data
- ADC failure

 Your panel (CRITICAL interpretation)

You have:

- **Hydraulic LOW PRESSURE (A & B)**
- **Standby hydraulic degraded**
- **Speed Trim FAIL**
- **Mach Trim FAIL**
- **Auto Slat FAIL**
- **Feel Diff Press**

👉 That tells you:

⚠️ **Mach Trim did NOT fail independently — it is LOST due to hydraulic failure**

 Failure chain (what's happening)

Hydraulic systems A & B lost



Stabilizer trim system loses power



Mach Trim cannot operate



MACH TRIM FAIL light ON

✈️ Operational effect

With **MACH TRIM FAIL**:

- ❌ No automatic compensation for Mach tuck
- At high speed:
 - Aircraft becomes **less stable**
 - Pilot must manually control pitch

👉 Aircraft is still controllable ✅

But:

⚠️ Stability is reduced at high Mach

🧠 Line maintenance mindset

If MACH TRIM FAIL alone:

👉 Check:

- Mach trim computer
- Stabilizer trim system
- Air data inputs

If MACH TRIM FAIL + HYD issues (your case):

👉 Root cause is:

✅ **Loss of hydraulic power**

🔗 Tie to related lights (important pattern)

Light	Why it's also on
SPEED TRIM FAIL	Same stabilizer dependency

AUTO SLAT FAIL	Slats require hydraulic power
FEEL DIFF PRESS	No hydraulic differential pressure
YAW DAMPER	Needs hydraulic B pressure

⚡ Simple way to remember

MACH TRIM FAIL = “no automatic high-speed pitch stability correction”

☑ Summary (your exact scenario)

- **MACH TRIM FAIL (amber)** = Mach trim system inoperative
- Root cause:
 -  Loss of Hyd System A
 -  Loss of Hyd System B
 -  Standby system degraded
- Result:
 - Reduced high-speed stability

💡 Big-picture (this entire panel)**

All these lights together show:

 **A complete hydraulic system failure causing cascade loss of flight control functions**

And:

 **MACH TRIM FAIL is a downstream symptom — not the primary failure**

☑ Final takeaway

When you see:

- **SPEED TRIM FAIL + MACH TRIM FAIL + FEEL DIFF PRESS + HYD LOW PRESSURE**

 Immediately think:

 **Hydraulic failure → not individual control system failures**

AUTO SLAT FAIL

AUTO SLAT FAIL Light (Amber) — What it means on the 737

Looking at your panel, the **AUTO SLAT FAIL light (amber)** is illuminated along with multiple flight control and hydraulic warnings.

Definition

The **AUTO SLAT FAIL light** indicates that the **automatic slat system is not operating or has been disabled**.

What the Auto Slat System does

The 737 **automatic slat system** provides **stall protection** by automatically extending slats when needed.

It works:

- **With flaps extended (especially flaps 1–5)**
- When the aircraft approaches a **high angle of attack (AOA)**

 If triggered, it moves:

- **Leading edge slats to FULL EXTEND**

Why this is important

The system:

- Prevents aerodynamic stall
- Improves lift at low speed
- Enhances safety margins during:
 - Takeoff
 - Approach

What triggers AUTO SLAT FAIL

1.  **Hydraulic pressure loss (MOST IMPORTANT in your case)**

Auto slats require:

- **Hydraulic System B** (primary source for leading edge devices)

👉 In your image:

- ☒ **System A LOW PRESSURE**
- ☒ **System B LOW PRESSURE**
- ☒ **Standby hydraulic degraded**

2. ⚙️ **Slat drive unit or mechanism failure**

- Mechanical issue
- Jammed system

3. ⚡ **Electrical/control system fault**

- Fault in slat control electronics
- Sensor or logic failure (AOA inputs)

4. 📄 **Invalid AOA or flight data**

- System cannot determine when to deploy

🔗 **Your panel (CRITICAL interpretation)**

You have ALL of these:

- **Hydraulic LOW PRESSURE (A & B)** ☒
- **Standby hydraulic issues** ☒
- **SPEED TRIM FAIL** ☒
- **MACH TRIM FAIL** ☒
- **FEEL DIFF PRESS** ☒
- **YAW DAMPER** ☒
- **AUTO SLAT FAIL** ☒

👉 That clearly shows:

⚠️ **AUTO SLAT FAIL is NOT an isolated failure**

👉 It's caused by **loss of hydraulic power**

Failure chain (what's happening)

Hydraulic System B lost



Leading edge slats lose hydraulic actuation



Auto slat system cannot move slats



AUTO SLAT FAIL light ON

Operational effect

With AUTO SLAT FAIL:

-  Automatic stall protection lost
- Slats:
 - Will NOT extend automatically
- Pilot must:
 - Rely on manual configuration

 Aircraft still flyable 

But:

 Reduced safety margin at low speeds

Line maintenance mindset

If AUTO SLAT FAIL alone:

 Check:

- Slat system
- AOA inputs
- Control logic

If AUTO SLAT FAIL + HYD LOW PRESSURE (your panel):

 Root cause is:

 **Hydraulic failure (especially System B)**

Connection with other lights (pattern recognition)

Light	Why it's ON
SPEED TRIM FAIL	Needs stabilizer/hydraulic support
MACH TRIM FAIL	Same stabilizer dependency
FEEL DIFF PRESS	Needs A & B pressure
YAW DAMPER	Uses hydraulic B
AUTO SLAT FAIL	Uses hydraulic B

Simple way to remember

AUTO SLAT FAIL = “no automatic slat deployment for stall protection”

Summary (your exact scenario)

- **AUTO SLAT FAIL (amber)** = auto slat system inoperative
- Root cause:
 -  Loss of Hydraulic System B (primary)
 -  Overall hydraulic system degradation
- Effect:
 - No automatic stall protection via slats

Big-picture insight (VERY important)

Your entire panel now shows a **classic cascade failure**:

 **Hydraulic failure** → **loss of flight control support systems** → **multiple FAIL lights (speed trim, mach trim, auto slats, feel system, yaw damper)**

Final takeaway

When you see this combination:

- **AUTO SLAT FAIL + SPEED TRIM FAIL + MACH TRIM FAIL + HYD LOW PRESSURE**

 Don't troubleshoot each system separately.

 Go straight to **hydraulic system failure** as the root cause

LOW QUANTITY LIGHT

LOW QUANTITY Light (Amber) — What it means on the 737

Looking at your image, the **LOW QUANTITY light (amber)** is illuminated in the **STANDBY HYD section**.

Definition

The **LOW QUANTITY light** indicates that the **hydraulic fluid level in the standby hydraulic reservoir is low**.

What system this applies to

This light is specifically for:  **Standby Hydraulic System**

The standby system is the **backup** for:

- Rudder (manual / standby control)
- Leading edge devices (in certain failures)

What causes LOW QUANTITY

1. **Hydraulic fluid leak (MOST IMPORTANT)**

- Fluid has left the system
- Comes from:
 - Lines
 - Actuators
 - Reservoir leaks

 This is the **#1 thing to suspect**

2. **Low servicing level**

- Reservoir not properly serviced
- Fluid below minimum

3. ⚙️ System usage after a failure

- Standby system activated:
 - Loss of System A/B forces fluid circulation
- If leak exists → quantity drops rapidly

🔧 Your panel (CRITICAL interpretation)

You have:

-  **LOW QUANTITY (standby)**
-  **LOW PRESSURE (A & B)**
-  **LOW PRESSURE (standby)**
-  Multiple flight control failures

👉 This tells you:

⚠️ **There is likely a hydraulic fluid loss affecting the entire aircraft**

📋 Failure chain (YOUR exact scenario)

Hydraulic fluid leak or loss



Reservoir levels drop



LOW QUANTITY (standby) light



Hydraulic pressure lost (A, B, standby)



Flight control systems fail



Multiple warnings (speed trim, mach trim, etc.)

🚨 Why this is critical

LOW QUANTITY is often the **ROOT CAUSE**, not just a symptom

Because:

- Pressure loss = result

- Fluid loss = source problem

Line maintenance mindset

If LOW QUANTITY alone:

 Check:

- Reservoir level
- Servicing

If LOW QUANTITY + LOW PRESSURE (your case):

 Immediately suspect:

 **Hydraulic leak**

What to inspect:

- Hydraulic reservoirs
- Lines along fuselage/wings
- Actuators (gear, flight controls)
- Evidence of fluid (wet areas, stains)

Operational impact

With LOW QUANTITY:

- System may:
 - Lose pressure completely
 - Overheat
 - Become unusable

 Leads to:

- Loss of backup hydraulic function

Important distinction

Light	Meaning
LOW PRESSURE	System not producing pressure

LOW QUANTITY	Not enough fluid in system 
OVERHEAT	Fluid too hot

👉 **LOW QUANTITY = supply problem (fluid gone)**

⚡ Simple way to remember

LOW QUANTITY = "you're running out of hydraulic fluid"

☑ Summary (your exact panel)

- **LOW QUANTITY (amber)** = standby hydraulic fluid level is low
- Combined with:
 - LOW PRESSURE (A, B, STBY)

👉 Confirms:

⚠ **Major hydraulic fluid loss or leak**

💡 Big-picture (this whole scenario)

From everything you've shown, this panel is essentially telling you:

⚠ **Hydraulic fluid is gone → pressure drops → flight control systems fail**

☑ Final AMT takeaway

When you see:

- **LOW QUANTITY + LOW PRESSURE (multiple systems)**

👉 Don't chase pumps first.

✅ Go straight to:

- **Hydraulic leak inspection**
- **Fluid quantity verification**

STBY RUD ON

▣ **STBY RUD ON Light (Amber) — What it means on the 737**

Looking at your panel, the **STBY RUD ON light (amber)** is illuminated in the **STANDBY HYD section**, along with:

- **LOW QUANTITY**
- **LOW PRESSURE**
- Multiple flight control failure lights

☑ **Definition**

The **STBY RUD ON light** indicates that the **standby hydraulic system is powering the rudder**.

🔑 **What the standby rudder system does**

The 737 has a **Standby Hydraulic System** that acts as a **backup** if normal hydraulics fail.

It can power:

-  **Rudder (primary function)**
-  Some leading edge devices (in certain failures)

⚠ **When STBY RUD ON comes on**

1. ☑ **Automatic activation (MOST IMPORTANT in your case)**

The system automatically engages when:

- **Both Hydraulic System A and B pressure are low**

👉 In your panel:

-  **A LOW PRESSURE**
-  **B LOW PRESSURE**

2. 🖐 **Manual activation**

- Crew places **STBY RUD switch to ON**

🔗 **Your panel (CRITICAL interpretation)**

You have:

-  **LOW PRESSURE (A & B)**
-  **STBY LOW PRESSURE**
-  **LOW QUANTITY (standby)**
-  **STBY RUD ON**
-  Multiple flight control failures

👉 This confirms:

⚠️ **The aircraft has lost normal hydraulic systems A and B, so it switched to standby rudder operation**

BUT...

👉 You also have:

- **STANDBY LOW PRESSURE**
- **LOW QUANTITY**

💡 **What that REALLY means (VERY important)**

Hydraulic A & B lost

↓

Standby system activates → STBY RUD ON

↓

But standby system also has:

- LOW QUANTITY
- LOW PRESSURE

↓

Backup system is ALSO failing

✨ **Key takeaway**

⚠️ **You don't just have a failure — you're losing BOTH normal AND backup hydraulic capability**

✈️ **Operational effect**

With **STBY RUD ON**:

-  Rudder is powered by standby system (initially)

BUT in your case:

- ❌ Standby pressure is also low

👉 Result:

- Rudder authority **limited or lost**
- Aircraft control becomes **severely degraded**

🧠 Line maintenance mindset

If STBY RUD ON alone:

👉 Normal response to:

- Loss of A or B (or both)

If STBY RUD ON + LOW PRESSURE + LOW QUANTITY (your panel):

👉 Indicates:

🚨 **Severe hydraulic fluid loss or system failure affecting ALL hydraulic systems**

🔑 Relationship with other lights

Light	What it means
STBY RUD ON	Backup rudder system active
LOW PRESSURE (A & B)	Primary hydraulics lost
LOW PRESSURE (STBY)	Backup system also failing
LOW QUANTITY	Fluid loss (likely root cause)

⚡ Simple way to remember

STBY RUD ON = “backup rudder is in use”

👉 But in your case:

“...and even the backup is failing”

☑ Summary (your exact scenario)

- **STBY RUD ON (amber)** = standby hydraulic system is powering the rudder

- Triggered by:
 -  Loss of Hyd System A
 -  Loss of Hyd System B
- Combined with:
 -  LOW PRESSURE (standby)
 -  LOW QUANTITY

👉 Final conclusion:

 **Complete hydraulic system degradation, including backup system**

💡 Big-picture (your entire panel)

Everything you've shown ties into one root problem:

 **Hydraulic fluid loss → pressure collapse → backup system activation → backup also fails**

 Final AMT insight

When you see:

- **STBY RUD ON + LOW PRESSURE + LOW QUANTITY**

👉 Don't troubleshoot individual components.

 Go straight to:

- **Hydraulic leak inspection**
- **Reservoir quantity**
- **System integrity**

BAT DISCHARGE LIGHT

 **BAT DISCHARGE Light (Amber) — What it means on the 737**

Looking at your image, the **BAT DISCHARGE light (amber)** is illuminated on the electrical panel, along with:

- **TR UNIT light**
- **ELEC light**
- **Standby Power OFF**

- DC indications on the meter

☑ Definition

The **BAT DISCHARGE light** indicates that the **battery is supplying power to the DC system (discharging) when it should not be.**

🔧 Normal operation

On a properly powered aircraft:

- Generators and TR units supply electricity
- Battery is:
 -  **Charged**
 -  **Standing by (not heavily discharging)**

👉 The BAT DISCHARGE light should be **OFF**

🔔 What BAT DISCHARGE means

When the light comes ON:

⚠ **The battery is providing power to the DC buses instead of being charged**

👉 This happens when:

- Normal power sources are **not supplying DC power**

⚠ Common causes

1. 🔌 **Loss of generator/AC power (MOST important in your panel)**

- Generators not online or not supplying loads
- Transfer buses unpowered

2. ⚡ **TR (Transformer Rectifier) unit failure**

- TR units convert AC → DC
- If they fail:
 - DC system loses normal power

- Battery steps in → discharge

👉 You also have a **TR UNIT light ON**, which supports this

3. 📄 **Battery powering essential buses**

- Occurs during:
 - Electrical failures
 - Emergency configurations

4. ⚙️ **Electrical distribution issue**

- Bus failure
- Power not reaching DC system

🔗 **Your panel (CRITICAL interpretation)**

You have:

- ✅ **BAT DISCHARGE**
- ✅ **TR UNIT light**
- ✅ **ELEC light**
- ✅ **Standby PWR OFF**
- ✅ Multiple system failures (from earlier panels: IRS, probe heat, EEC, etc.)

👉 This clearly shows:

Loss of AC power (generators / buses)

↓

TR units stop supplying DC power

↓

DC buses lose normal power

↓

Battery takes over → discharging

↓

BAT DISCHARGE light ON

✦ Big conclusion (VERY important)

⚠️ This is NOT a battery problem — it is a major electrical power failure

The battery is: 👉 doing its job as backup

✈️ Operational effect

With BAT DISCHARGE:

- Battery:
 - Is being drained
- If not restored:
 - Time-limited electrical power

👉 Eventually:

- Loss of essential systems

🧠 Line maintenance mindset

If BAT DISCHARGE alone:

👉 Check:

- TR units
- DC system

If BAT DISCHARGE + TR UNIT + ELEC (your case):

👉 Go directly to:

✅ AC electrical system problem:

- Generators
- Transfer buses
- Bus tie logic

🔗 Tie to ALL your previous lights

You've shown:

- IRS ON DC / DC FAIL ✅
- GPS failure ✅

- Probe heat failure ✓
- Engine control ALTN ✓

👉 ALL of those depend on:

AC power feeding the airplane

⚡ Simple way to remember

BAT DISCHARGE = “battery is powering the airplane because normal power is gone”

☑ Summary

- **BAT DISCHARGE (amber)** = battery supplying DC power
- Causes:
 - Generator loss ✓
 - TR failure ✓
 - Electrical bus failure ✓
- In YOUR panel: 👉 Confirms **major electrical system failure (not a battery issue)**

💡 Final AMT-level takeaway

When you see:

- **BAT DISCHARGE + TR UNIT + ELEC + widespread system failures**

👉 Don't troubleshoot components individually.

✓ Go straight to:

1. **Generators (GEN 1 / GEN 2 / APU GEN)**
2. **AC transfer buses**
3. **Bus tie system**
4. **TR units (secondary check)**

TR UNIT

TR UNIT Light (Amber) — What it means on the 737

Looking at your image, the **TR UNIT light (amber)** is illuminated in the electrical panel, along with:

- **BAT DISCHARGE**
- **ELEC light**
- **Standby PWR OFF**
- Low/abnormal DC indications on the meter

Definition

The **TR UNIT light** indicates a **failure or malfunction in a Transformer Rectifier (TR) unit**.

What a TR Unit does

The 737 uses **Transformer Rectifiers (TR1, TR2, TR3)** to:

- ✓ Convert **AC power** → **DC power**

These TR units supply power to:

- **DC buses**
- Battery charging
- Many aircraft systems (IRS backup, instruments, controls, etc.)

What triggers the TR UNIT light

1. **TR unit failure (most direct cause)**

- TR unit stops converting AC to DC
- No output or abnormal output

2. **Loss of AC input power (VERY IMPORTANT in your case)**

- If the TR doesn't get AC power:
 - It can't produce DC
 - Light comes ON

 This is likely in your panel

3. ⚙️ Internal malfunction

- Overheat or internal fault
- Voltage/current out of limits

🔧 Your panel (CRITICAL interpretation)

You have:

- ✅ TR UNIT light
- ✅ BAT DISCHARGE
- ✅ ELEC light
- ✅ STANDBY PWR OFF
- ✅ Abnormal DC readings
- ✅ Numerous system failures across the aircraft (from previous panels)

📋 What's actually happening (chain)

Loss of AC electrical power (generators / buses)

↓

TR units lose AC input

↓

TR units stop producing DC power

↓

TR UNIT light illuminates

↓

DC buses lose normal power

↓

Battery takes over → BAT DISCHARGE

↓

Multiple systems fail

🔔 Key insight (VERY important)

⚠️ The TR UNIT light does NOT necessarily mean the TR itself failed

👉 It often means:

The TR has no AC power to work with

Line maintenance mindset

If TR UNIT light alone:

👉 Check:

- TR unit itself
- Output voltage

If TR UNIT + BAT DISCHARGE + ELEC (your case):

👉 Immediately think:

✅ **Upstream AC power failure**

Check:

1. Generators (GEN 1 / GEN 2)
2. APU GEN
3. AC transfer buses
4. Bus tie configuration

NOT:

- Replace TR first

Tie to your entire scenario (big picture)

Across all your questions you've shown:

- IRS ON DC / DC FAIL ✅
- GPS light ✅
- Probe heat failures ✅
- Engine control ALTN ✅
- PSEU ✅
- Hydraulic degradations ✅

👉 All of those systems depend on:

✅ **AC → converted to DC by TR units**

⚡ Simple way to remember

TR UNIT = “AC-to-DC converter not supplying DC power”

👉 But in your case:

“because AC power is missing”

☑ Summary

- **TR UNIT (amber)** = TR not supplying DC power
- Causes:
 - TR failure ❌ (possible but less likely here)
 - ✅ Loss of AC input power (most likely in your panel)
- Result:
 - DC system goes offline
 - Battery starts discharging

💡 Final AMT-level takeaway

When you see:

- **TR UNIT + BAT DISCHARGE + widespread aircraft system issues**

👉 Always diagnose in this order:

1. ✅ **AC power (generators / APU)**
2. ✅ **Transfer buses**
3. ✅ **Bus tie system**
4. THEN check TR units

💧 Bottom line for your panel

⚠ **TR UNIT light = symptom of a major electrical power failure, not the root cause**

ELEC

▨ ELEC Light (Amber) — What it means on the 737

Looking at your image, the **ELEC light (amber)** is illuminated on the electrical panel, along with:

-  **BAT DISCHARGE**
-  **TR UNIT**
-  **STANDBY PWR OFF**
-  Abnormal DC indications
-  Many aircraft systems failing across your earlier panels

Definition

The **ELEC light** indicates a **fault in the aircraft's electrical power system**.

What the ELEC light represents

The ELEC light is a **general electrical system warning** that tells you:

 **Something is wrong with normal electrical power supply or distribution**

It is **not tied to just one component** — it's a **system-level warning**.

What can cause the ELEC light

1. **Loss of generator power (MOST COMMON)**

- Engine generators not supplying power
- APU generator not supplying power

2. **AC transfer bus failure**

- Power not reaching the aircraft buses
- Bus tie system malfunction

3. **Electrical distribution problem**

- Power available, but not routed correctly

4. **Multiple electrical component failures**

- TR unit problems
- DC system issues

🔗 Your panel (CRITICAL interpretation)

You have this combination:

- ☒ **ELEC light**
- ☒ **TR UNIT**
- ☒ **BAT DISCHARGE**
- ☒ **STANDBY PWR OFF**
- ☒ IRS failures (ON DC, DC FAIL, FAULT)
- ☒ Probe heat failures
- ☒ Engine control in ALTN

📋 What's happening (full chain)

Loss of AC electrical power (generators / transfer buses)

↓

AC buses no longer powered

↓

TR units lose AC input → TR UNIT light

↓

DC system loses normal power

↓

Battery takes over → BAT DISCHARGE

↓

IRS switches to DC → ON DC

↓

DC also insufficient → DC FAIL → FAULT

↓

Multiple aircraft systems lose power

↓

ELEC light illuminates (system-level fault)

🔔 Key takeaway (VERY important)

☒ **ELEC light = BIG PICTURE problem**

👉 Not a single component failure

Line maintenance mindset

If ELEC light alone:

👉 Check:

- Generators
- Electrical buses

If ELEC + TR UNIT + BAT DISCHARGE + system-wide failures (your case):

👉 Go directly to:

✅ **Primary electrical sources**

1. Engine generators (GEN 1 / GEN 2)
2. APU generator
3. External power

✅ Then: 4. AC transfer buses

5. Bus tie system

Tie to ALL your previous lights

Everything you've asked about connects to this:

- IRS ON DC / DC FAIL ✅
- GPS loss ✅
- Probe heat failures ✅
- Engine control ALTN ✅
- PSEU ✅

👉 ALL rely on:

Stable AC electrical power

Simple way to remember

ELEC light = "the airplane lost normal electrical power"

☑ **Summary**

- **ELEC (amber)** = electrical system fault

- Most common cause:
 - ☒ Loss of generator power
 - ☒ AC bus failure
- In your scenario: 🖐️ **Confirms major electrical failure affecting the entire aircraft**

💡 Final AMT-level insight

When you see:

- **ELEC + BAT DISCHARGE + TR UNIT + widespread system failures**

🖐️ You are NOT dealing with multiple failures

⚠️ **You are dealing with ONE root problem: electrical power loss**

💧 Bottom line (your full situation)

Between:

- Electrical failures ☒
- Hydraulic failures ☒ (likely independent or secondary)
- Bleed/pressurization issues ☒

🖐️ The biggest consistent pattern is:

⚠️ **Loss of electrical power → cascading system failures across the aircraft**

STANDBY PWR OFF

📺 **STANDBY POWER OFF Light (Amber) — What it means on the 737**

Looking at your panel, the **STANDBY PWR OFF light (amber)** is illuminated along with:

- ☒ **ELEC light**
- ☒ **BAT DISCHARGE**
- ☒ **TR UNIT**
- ☒ **IRS ON DC / DC FAIL / FAULT**
- ☒ Multiple system failures across the aircraft

Definition

The **STANDBY PWR OFF light** indicates that the **standby electrical power system is NOT supplying power to the standby (essential) buses when it should be.**

What the Standby Power System does

The **standby power system** is your **last line of electrical backup.**

It normally powers:

-  Standby attitude indicator
-  Essential flight instruments
-  Some navigation equipment
-  Critical DC/AC standby buses

Normal operation

- With normal AC power available: → Standby system is **armed (ready)**
→ **STANDBY PWR OFF light = OFF**
- If AC power is lost: → System automatically switches to:
 - Battery
 - Standby inverter

What STANDBY PWR OFF means

When the light comes ON:

 **The standby buses are NOT being powered when they should be**

Common causes

1. Loss of AC power (PRIMARY cause in your case)

- Generators offline
- Transfer buses unpowered

2. 📄 **Battery/standby system not supplying power**

- Battery depleted or disconnected
- Standby inverter failure

3. ⚡ **Electrical distribution failure**

- Power not reaching standby buses
- Bus switching logic failed

4. ⚙️ **Standby power switch in OFF (less likely here)**

- If switch is set to OFF, system won't power buses

🔗 **Your panel (CRITICAL interpretation)**

You have:

- ✅ **STANDBY PWR OFF**
- ✅ **BAT DISCHARGE**
- ✅ **TR UNIT**
- ✅ **ELEC**
- ✅ IRS failures (ON DC → DC FAIL → FAULT)
- ✅ Multiple unrelated systems failing

📋 **Full failure chain (this is the key)**

Loss of AC electrical power (generators / transfer buses)

↓

AC buses go offline

↓

TR units stop producing DC → TR UNIT light

↓

Battery takes over → BAT DISCHARGE

↓

Standby system should take over...

↓

BUT standby power not supplying buses

↓

STANDBY PWR OFF light ON

↓

IRS and critical systems lose power



Widespread aircraft system failures

✦ What this REALLY means

⚠️ **You are losing BOTH normal AND standby electrical power**

This is extremely important:

- Normal power ✅ lost
- Backup power ⚠️ not available

👉 That's why you see:

- IRS failures
- Probe heat failures
- Engine control ALTN
- GPS loss

🗣️ Line maintenance mindset

If STANDBY PWR OFF alone:

👉 Check:

- Battery
- Standby inverter
- Standby power switch

If STANDBY PWR OFF + ELEC + BAT DISCHARGE (your case):

👉 Go directly to:

✅ Electrical system root cause

1. Generators (GEN 1 / GEN 2 / APU GEN)
2. AC transfer buses
3. Bus tie system
4. DC distribution

Important distinction

Light	Meaning
ELEC	Electrical system fault (big picture)
TR UNIT	AC → DC conversion lost
BAT DISCHARGE	Battery supplying power
STBY PWR OFF	Backup system not powering buses 🚨

⚡ Simple way to remember

STBY PWR OFF = “even the backup electrical system is not working”

☑ Summary

- **STANDBY PWR OFF (amber)** = standby electrical buses not powered
- Causes:
 - AC power loss ✅
 - Battery / inverter issue ✅
 - Electrical distribution failure ✅

💡 Big-picture conclusion (your full scenario)

Across EVERYTHING you’ve shown:

⚠ **This is a major electrical power failure affecting the entire aircraft**

And specifically:

! **STANDBY PWR OFF confirms the backup system is NOT saving you**

💧 Final AMT takeaway

When you see:

- **ELEC + TR UNIT + BAT DISCHARGE + STBY PWR OFF**

👉 Immediately think:

✅ **Total electrical system degradation (primary + standby)**

DRIVE LIGHT

DRIVE Light (Amber) — What it means on the 737

Looking at your panel, the **DRIVE lights (amber)** are illuminated for both:

- **Engine 1 IDG**
- **Engine 2 IDG**

Definition

The **DRIVE light** indicates a **problem with the Integrated Drive Generator (IDG) system**—specifically the **constant speed drive (CSD) portion** of the generator.

What the IDG / Drive does

Each engine has an **IDG (Integrated Drive Generator)** that:

- Produces **AC electrical power**
- Maintains **constant frequency (400 Hz)** using:
 - A **mechanical drive (CSD)**

 This is critical for powering:

- The entire aircraft electrical system
- AC transfer buses
- TR units → DC power

What triggers the DRIVE light

1. **IDG oil overtemperature (MOST COMMON)**

- The CSD uses oil for:
 - Cooling
 - Lubrication
- If oil overheats: → **DRIVE light ON**

2. 💧 **Low IDG oil quantity**

- Loss of oil
→ Overheat or damage

3. ⚙️ **CSD mechanical problem**

- Drive not maintaining constant speed
- Internal failure

4. 🔌 **Loss of proper generator drive**

- Engine may be running, but: → Generator not producing correct output

📋 **System behavior (IMPORTANT)**

When DRIVE light comes on:

⚠️ The IDG is considered **unreliable or damaged**

👉 Procedure (operationally):

- **IDG should be disconnected**

🔗 **Your panel (CRITICAL insight)**

You have:

- ☒ **DRIVE lights (both engines)**
- ☒ **ELEC light**
- ☒ **TR UNIT**
- ☒ **BAT DISCHARGE**
- ☒ **STANDBY PWR OFF**
- ☒ IRS failures (ON DC → DC FAIL → FAULT)
- ☒ Multiple aircraft systems failing

🔗 **This ties EVERYTHING together**

👉 This is the missing link:

IDG / DRIVE problem (both engines)

↓

Generators not supplying proper AC power
↓
AC transfer buses lose power
↓
TR units lose AC input → TR UNIT light
↓
DC buses lose power
↓
Battery takes over → BAT DISCHARGE
↓
Standby system not sustaining → STBY PWR OFF
↓
IRS goes ON DC → DC FAIL → FAULT
↓
Aircraft-wide system failures

✦ BIG CONCLUSION

⚠️ **The DRIVE lights explain WHY you lost electrical power**

This is likely:

👉 **Loss of both engine-driven generators (IDGs)**

🧠 Line maintenance mindset

If ONE DRIVE light:

👉 Problem with one IDG

If BOTH DRIVE lights (your case):

👉 Very important:

⚠️ **Common cause affecting BOTH IDGs OR both generators taken offline**

Possible causes:

- IDG overheating (both)
- Oil system issue
- Generator disengagement

- Or crew disconnected both IDGs

✈ Operational effect

With both DRIVE lights:

- ✖ Both engine generators affected
- ✖ Loss of normal AC power
- ✖ Aircraft runs on:
 - APU GEN (if available)
 - OR battery

🔑 Important distinction

Light	Meaning
DRIVE	Problem with generator drive system
GEN OFF BUS	Generator not connected to bus
ELEC	Electrical system fault
BAT DISCHARGE	Battery supplying power

⚡ Simple way to remember

DRIVE light = “the generator drive system is overheating or failing”

☑ Summary

- **DRIVE (amber)** = IDG/CSD problem
- Causes:
 - Overheat ✔
 - Oil issue ✔
 - Mechanical failure ✔
- In YOUR panel: 🖐 **Both DRIVE lights = loss of both engine generators → total electrical failure**

💡 FINAL BIG-PICTURE (everything you asked about)

You’ve basically built a **complete failure scenario**:

⚠️ **Both IDGs fail or are lost (DRIVE lights)** → AC power lost

- TR units fail
- Battery discharges
- Standby power insufficient
- IRS + avionics fail
- Aircraft systems degrade

💧 **Final AMT takeaway**

When you see:

- ☒ **BOTH DRIVE lights**
- ☒ **BAT DISCHARGE**
- ☒ **TR UNIT**
- ☒ **ELEC**
- ☒ **STANDBY PWR OFF**

👉 You can confidently say:

Root cause = loss of engine-driven electrical power (IDGs)

EQUIP COOLING OFF LIGHT

📺 **EQUIP COOLING OFF Light (Amber) — What it means on the 737**

Looking at your image (top right of the panel), both **EQUIP COOLING “OFF” lights (amber)** are illuminated for:

- ☒ **SUPPLY fan = OFF**
- ☒ **EXHAUST fan = OFF**

☑️ **Definition**

The **EQUIP COOLING OFF light** indicates that the **equipment cooling fan (supply or exhaust) is NOT operating**.

🔧 **What the equipment cooling system does**

The **equipment cooling system** keeps the **E/E (electronics compartment)** cool.

It uses:

-  **Supply fan** → pushes cool air into avionics
-  **Exhaust fan** → pulls hot air out

👉 This cools:

- FMCs
- Displays
- IRSs
- Electrical equipment

Why this matters

Without cooling → avionics can **overheat very quickly**

What causes EQUIP COOLING OFF lights

1. **Loss of electrical power (MOST IMPORTANT in your case)**

Fans are electrically powered.

If you lose:

- AC power
- DC power

👉 Fans stop → lights come ON

2. **Fan failure**

- Supply fan failed
- Exhaust fan failed

3. **Switch in ALTN / OFF configuration**

- System can run in alternate mode
- But if no power → both off

4. **Airflow restriction (less common)**

- Duct blockage

🔧 Your panel (CRITICAL interpretation)

You have:

- ☒ **EQUIP COOLING OFF (both fans)**
- ☒ **ELEC light**
- ☒ **TR UNIT**
- ☒ **BAT DISCHARGE**
- ☒ **STANDBY PWR OFF**
- ☒ **DRIVE lights (both engines)**
- ☒ IRS failures (ON DC → DC FAIL → FAULT)

🗨️ What's happening (full chain)

Both IDGs (generators) lost → DRIVE lights



AC electrical power lost



Fans lose electrical power



Supply & exhaust fans stop



EQUIP COOLING OFF lights ON



Avionics lose cooling



Avionics begin failing (IRS, GPS, FMC, etc.)

🌟 Key takeaway (VERY important)

⚠️ **EQUIP COOLING OFF is NOT the root problem** 🙌 It is a **result of electrical power loss**

🧠 Line maintenance mindset

If ONE fan OFF:

🙌 Likely fan failure

If BOTH fans OFF (your case):

👉 Think immediately:

✅ **Electrical power loss**

Do NOT troubleshoot fans first

🔗 **Tie to the EQUIP (smoke) light you asked earlier**

- **EQUIP COOLING OFF** → **avionics overheat**
- If severe: → could eventually trigger **EQUIP (smoke) warning**

✈️ **Operational effect**

With both fans OFF:

- ❌ No cooling in E/E bay
- ❌ Avionics overheating risk
- ❌ Navigation + systems degrade

🔑 **Simple way to remember**

EQUIP COOLING OFF = “avionics cooling airflow has stopped”

But in your case:

“...because electrical power is gone”

☑️ **Summary (your exact scenario)**

- **SUPPLY OFF + EXHAUST OFF** = both cooling fans not running
- Most likely cause:
 - ✅ **Loss of electrical power**
- Result:
 - Avionics lose cooling
 - Contributes to system failures

💡 **BIG FINAL PICTURE (your entire aircraft scenario)**

You’ve now built a **complete system failure chain**:

⚠️ **Both IDGs (DRIVE lights) fail or are disconnected** → AC power lost

- TR units fail
- Battery discharging
- Standby power not supplying
- Cooling fans shut off
- Avionics overheat / fail
- IRS / GPS / probe heat / EEC issues

💧 **Final AMT takeaway**

When you see:

- **EQUIP COOLING OFF (both)**
- **ELEC + TR UNIT + BAT DISCHARGE**
- **DRIVE lights**

👉 Diagnosis:

✅ **Loss of AC electrical power — NOT cooling system failure**

NOT ARMED

📺 **NOT ARMED Light (Amber) — What it means on the 737**

Looking at your image (right side panel under **EMER EXIT LIGHTS**), the **NOT ARMED light (amber)** is illuminated.

☑️ **Definition**

The **NOT ARMED light** indicates that the **Emergency Exit Lighting system is NOT armed for automatic operation.**

🔑 **What the Emergency Exit Lighting system does**

This system provides:

- ✅ **Floor path lighting**
- ✅ **Exit signs**
- ✅ **Emergency cabin lighting**

👉 It comes on automatically during:

- Loss of electrical power
- Emergency situations (evacuation)

⚙️ **Normal configuration**

The switch has positions:

- **ON** → lights ON manually
- **ARMED** → lights activate automatically ✅
- **OFF** → lights OFF (not protected)

⚠️ **What triggers the NOT ARMED light**

1. ❌ **Switch NOT in ARMED (MOST COMMON)**

- If switch is:
 - **OFF**
 - OR not properly set to **ARMED**

👉 The light comes on

2. ⚙️ **System not armed electrically**

- Even if switch is set correctly, but:
 - No power to system
 - Fault in circuitry

3. 🔋 **Emergency lighting battery issue**

- Internal battery not ready or faulted

🔗 **Your image (CRITICAL observation)**

From the image:

- The **EMER EXIT LIGHTS** selector is positioned at **OFF**
- The **NOT ARMED** light is illuminated

👉 That means:

✅ **This is a NORMAL indication for this configuration**

Because:

Switch in OFF

↓

System not armed

↓

NOT ARMED light ON

⚠️ Why this matters

If the system is NOT ARMED:

- ❌ Emergency lights will NOT activate automatically
- ❌ In a power loss → cabin could go dark

🗨️ Line maintenance / operational mindset

If NOT ARMED + switch OFF:

✅ Normal indication

If NOT ARMED + switch appears ARMED:

👉 Then suspect:

- Electrical issue
- Battery problem
- Control circuit fault

🔑 Important distinction

Switch Position	Light	Meaning
ARMED	OFF	✅ System ready
OFF	ON	⚠️ Not armed
ON	OFF	Lights manually on

⚡ Simple way to remember

NOT ARMED = “emergency lights won’t come on automatically”

☑ Summary

- **NOT ARMED (amber)** = emergency exit lighting system not armed
- In your image:
 -  Switch is OFF
 -  Light is ON

👉 Conclusion:

✓ **Normal indication — system intentionally not armed**

💡 Big-picture tie-in

Unlike many of your other lights:

- This is **NOT part of the electrical failure chain**
- It is simply a **configuration status**

☑ Final takeaway

NOT ARMED light = **configuration warning, not a system failure (unless switch is already ARMED)**

TRANSFER BUSS OFF LIGHT

▨ **TRANSFER BUS OFF Light (Amber) — What it means on the 737**

Looking at your image (center electrical panel), you have:

-  **Left TRANSFER BUS OFF**
-  **Right TRANSFER BUS OFF**
-  **SOURCE OFF**
-  **GEN OFF BUS**
-  **APU GEN OFF BUS**

Definition

The **TRANSFER BUS OFF light** indicates that the **corresponding AC Transfer Bus is NOT being powered**.

What a Transfer Bus does

The 737 has:

- **Transfer Bus 1**
- **Transfer Bus 2**

These buses distribute **AC electrical power** to almost all major aircraft systems.

Power normally comes from:

-  Engine generators (IDGs)
-  APU generator
-  External power

Normal condition

- Transfer buses are powered
-  **TRANSFER BUS OFF lights = OFF**

What TRANSFER BUS OFF means

When the light is ON:

 **That AC transfer bus has NO electrical power**

What causes TRANSFER BUS OFF

1. **Loss of generator power (PRIMARY cause in your case)**

- Engine generators not supplying power
- APU generator not supplying power

2. **Bus tie / transfer failure**

- Power exists, but not routed to the bus

3. ⚙️ **Bus transfer switch OFF**

- Prevents automatic power routing

(In your image, **BUS TRANSFER switch is OFF** — important)

4. ⚡ **Electrical fault in distribution system**

- Bus not receiving power due to system fault

🔗 **Your image (CRITICAL analysis)**

You have ALL of these:

- ✅ **Both TRANSFER BUS OFF lights**
- ✅ **SOURCE OFF**
- ✅ **GEN OFF BUS (both engines)**
- ✅ **APU GEN OFF BUS**
- ✅ **DRIVE lights (both engines)**
- ✅ **BAT DISCHARGE**
- ✅ **TR UNIT**
- ✅ **ELEC**
- ✅ **STANDBY PWR OFF**
- ✅ **EQUIP COOLING OFF**

📋 **Complete failure chain (this is the key)**

Both IDGs (engine generators) unavailable → DRIVE lights

↓

No AC power from engines

↓

APU generator not supplying power

↓

No AC source available

↓

Both transfer buses lose power

↓

TRANSFER BUS OFF lights ON (both sides)

↓

TR units lose AC input → TR UNIT light

↓

DC system loses power



Battery takes over → BAT DISCHARGE



Standby system fails → STBY PWR OFF



Aircraft systems fail (IRS, probe heat, EEC, etc.)

✦ **BIG takeaway (VERY important)**

⚠ **TRANSFER BUS OFF = the airplane has lost AC electrical power distribution**

And in your case:

! **BOTH buses are OFF → total AC power loss**

🧠 **Line maintenance mindset**

If ONE TRANSFER BUS OFF:

👉 Likely:

- Single generator or bus issue

If BOTH TRANSFER BUS OFF (your case):

👉 Immediately think:

✅ **NO AC power source is supplying the airplane**

Check:

1. Generators (IDGs)
2. APU GEN
3. External power
4. Bus transfer switch

🔑 **Important relationships**

Light	Meaning
TRANSFER BUS OFF	AC bus not powered

GEN OFF BUS	Generator not connected to bus
SOURCE OFF	No power source selected
DRIVE	Generator drive problem (root cause)
TR UNIT	No AC → no DC conversion

⚡ Simple way to remember

TRANSFER BUS OFF = “this side of the airplane lost AC power”

Both ON = “the airplane lost ALL AC power”

☑ Summary (your exact scenario)

-  Both **TRANSFER BUS OFF** lights **ON**
-  No generators powering buses
-  No APU power
-  Electrical system running on battery (temporarily)

👉 Final conclusion:

⚠ **Total AC electrical power loss due to loss of generators (confirmed by DRIVE lights)**

💡 Final big-picture conclusion (everything you’ve shown)

You’ve built a complete scenario:

⚠ **Both IDGs fail or are disconnected (DRIVE lights)**

- No AC power
- Transfer buses go OFF
- TR units fail
- Battery discharges
- Standby power not supplying
- Cooling lost
- Avionics fail

💧 Final AMT takeaway

When you see:

- **BOTH TRANSFER BUS OFF**

- **GEN OFF BUS**
- **DRIVE lights**
- **BAT DISCHARGE**

👉 You can confidently diagnose:

✅ **Complete loss of AC electrical generation and distribution**

SOURCE OFF LIGHT

📺 **SOURCE OFF Light (Amber) — What it means on the 737**

Looking at your image, both **SOURCE OFF lights (amber)** are illuminated—one for each side of the electrical system.

✅ **Definition**

The **SOURCE OFF light** indicates that the **associated transfer bus does NOT have a valid electrical power source feeding it.**

🔑 **What “source” means**

A “source” is any **AC power supply feeding the transfer bus**, such as:

- ✅ Engine Generator (IDG)
- ✅ APU Generator
- ✅ External (ground) power

⚠️ **What SOURCE OFF is telling you**

When the light is ON:

⚠️ **No valid power source is available to feed that transfer bus**

👉 Not just disconnected —

👉 but **nothing available to power it**

Your image (VERY IMPORTANT context)

On your panel, you have:

- ☒ **SOURCE OFF (both sides)**
- ☒ **TRANSFER BUS OFF (both sides)**
- ☒ **GEN OFF BUS (both engines)**
- ☒ **APU GEN OFF BUS**
- ☒ **DRIVE lights (both engines)**
- ☒ **BAT DISCHARGE**
- ☒ **TR UNIT**
- ☒ **ELEC**
- ☒ **STANDBY PWR OFF**

What's happening (clean chain)

Both IDGs unavailable → DRIVE lights

↓

No engine generator power

↓

APU generator not supplying power

↓

No AC source available at all

↓

SOURCE OFF lights ON

↓

Transfer buses not powered → TRANSFER BUS OFF

↓

TR units lose AC input → TR UNIT light

↓

DC buses lose power

↓

Battery takes over → BAT DISCHARGE

↓

Standby system fails → STBY PWR OFF

↓

Aircraft systems fail

✦ **Key meaning (this is important)**

✅ **SOURCE OFF = “there is NO electrical power source available”**

🧠 **Line maintenance mindset**

If SOURCE OFF (one side):

👉 Check:

- That side generator
- Bus connection

If SOURCE OFF (both sides) — YOUR CASE:

👉 Immediately think:

⚠️ **Total loss of all AC power sources**

Check in this order:

1. Engine generators (IDGs)
2. APU generator
3. External power connection
4. Generator control / disconnect status

🔑 **Relationship with other lights**

Light	Meaning
SOURCE OFF	No power source available !
GEN OFF BUS	Generator not connected
TRANSFER BUS OFF	Bus not powered
DRIVE	Generator drive problem (root cause)

⚡ **Simple way to remember**

SOURCE OFF = “nothing is feeding the bus”

☑ Summary (your exact scenario)

- ✅ Both **SOURCE OFF** lights **ON**
- ✅ No generator power
- ✅ No APU power
- ✅ No external power selected

👉 Final conclusion:

⚠ **The aircraft has ZERO available AC electrical sources**

💡 **BIG PICTURE** (everything you've built)

All your electrical lights now line up perfectly:

⚠ **Both IDGs lost (DRIVE lights)**

- No AC generation
- SOURCE OFF
- TRANSFER BUS OFF
- TR units fail
- Battery discharges
- Standby power fails
- Aircraft-wide system failure

💧 **Final AMT takeaway**

When you see:

- **SOURCE OFF + TRANSFER BUS OFF + GEN OFF BUS + DRIVE lights**

👉 You can confidently diagnose:

✅ **Total loss of electrical generation (no AC power source available)**

FILTER BYPASS

📺 **FILTER BYPASS Light (Amber) — What it means on the 737**

Looking at your image (fuel panel lower left), both **FILTER BYPASS lights (amber)** are illuminated on the **left and right engine fuel sections**, along with:

-  **LOW PRESSURE** lights on fuel pumps
-  **FUEL TEMP** gauge visible
-  Fuel system controls (CROSSFEED, pumps, etc.)

Definition

The **FILTER BYPASS light** indicates that the **engine fuel filter has become restricted, and fuel is bypassing the filter to keep the engine running.**

What the fuel filter does

Each engine has a **fuel filter** to:

- Remove contaminants (debris, ice crystals, dirt)

Normally:

- Fuel flows **through the filter** before reaching the engine

What “bypass” means

If the filter becomes:

- Blocked
- Restricted

 The system opens a **bypass valve**

So fuel:

-  Does NOT pass through the filter
-  Goes directly to the engine (unfiltered)

Important meaning

 **FILTER BYPASS = the filter is clogged or restricted**

BUT:

- Engine still gets fuel 
- However:
 - Fuel is **no longer being filtered**

Common causes

1. ❄️ Fuel icing (VERY common)

- Ice forms in the filter
- Restricts flow → bypass activates

2. 🧴 Contamination

- Dirt, debris, microbial growth in fuel

3. 🔑 Cold fuel temperature

- Your panel shows a fuel temp gauge → important clue
- Cold fuel increases icing risk

4. ⚙️ Filter element blockage

- Normal wear or contamination buildup

Your panel interpretation (VERY important)

You have:

-  **FILTER BYPASS (both engines)**
-  **LOW PRESSURE (fuel pumps)**
-  Electrical system failure (from earlier):
 - DRIVE lights
 - TRANSFER BUS OFF
 - BAT DISCHARGE

 What this suggests:

 This is likely **NOT just two clogged filters at once**

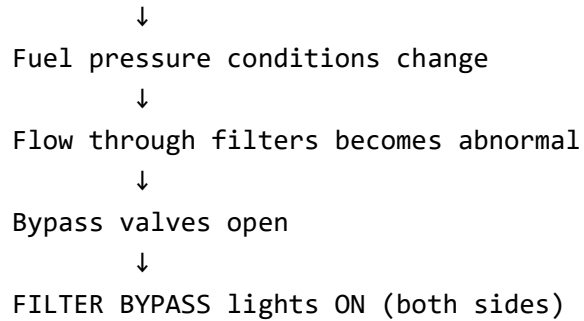
Instead:

Most likely scenario:

Electrical power lost



Fuel boost pumps not operating normally



Line maintenance mindset

If ONE FILTER BYPASS:

👉 Likely:

- Contamination
- Ice

If BOTH FILTER BYPASS (your case):

👉 Strong clue:

✅ **System-wide issue (fuel, electrical, or temperature related)**

Tie to other lights on your panel

Light	Relationship
LOW PRESSURE (fuel pumps)	Pumps not supplying proper pressure
FILTER BYPASS	Flow restriction or abnormal flow
ELEC / DRIVE / TR UNIT	Electrical failure affecting pumps

Operational effect

- Engine continues running ✅
- BUT:
 - ❌ Fuel is **unfiltered**
 - Risk of:
 - Contaminant ingestion
 - Engine damage (long-term)

Important distinction

Light	Meaning
FILTER BYPASS	Fuel bypassing clogged filter
LOW PRESSURE	Not enough fuel pressure
FUEL TEMP	Temperature affecting icing risk

Simple way to remember

FILTER BYPASS = “fuel is skipping the filter because it’s blocked”

Summary (your exact scenario)

- **Both FILTER BYPASS lights ON**
- Combined with:
 - Fuel pump LOW PRESSURE
 - Major electrical system issues

 Conclusion:

⚠ Fuel filters are being bypassed due to abnormal fuel system conditions — likely influenced by the larger electrical failure affecting pump operation

Big-picture connection

This fits into your overall aircraft failure:

⚠ Electrical failure → fuel pumps degraded → abnormal fuel flow → filter bypass

Final AMT takeaway

When you see:

- **BOTH FILTER BYPASS lights**

 Ask:

1. Is fuel contaminated or iced?
2. Are pumps operating normally?  (in your case: **no**)
3. Is there a larger system problem? 

ENG VALVE CLOSED LIGHT

ENG VALVE CLOSED Light (Blue) — What it means on the 737

Looking at your image, you have:

-  **ENG VALVE CLOSED (blue)**
-  **SPAR VALVE CLOSED (blue)**
-  **FILTER BYPASS (amber)**
-  Fuel system indications active

Definition

The **ENG VALVE CLOSED light** indicates that the **engine fuel shutoff valve (engine fuel valve)** is **CLOSED** — no fuel is flowing into that engine.

What the engine fuel valve does

Each engine has a **fuel shutoff valve (engine valve)** that:

- Controls fuel flow from the wing tank to the engine
- Is operated by the **engine start lever (fuel control lever)**

Normal operation

Engine running:

- Start lever → **IDLE**
- Valve → **OPEN**
-  **ENG VALVE CLOSED light OFF**

Engine shut down:

- Start lever → **CUTOFF**
- Valve → **CLOSED**
-  **ENG VALVE CLOSED light ON**

⚠️ What the ENG VALVE CLOSED light means

- Fuel is NOT being supplied to the engine

🔧 Your image (IMPORTANT context)

You have:

- ☒ **ENG VALVE CLOSED (both engines)**
- ☒ **SPAR VALVE CLOSED**
- ☒ **FILTER BYPASS**
- ☒ **LOW PRESSURE (fuel pumps)**
- ☒ Major **electrical system failure (from the rest of your panel):**
 - DRIVE lights
 - TRANSFER BUS OFF
 - SOURCE OFF
 - BAT DISCHARGE

📋 Interpreting YOUR scenario

Two possibilities:

☒ 1. Normal (most likely if engines are OFF)

If this is:

- Aircraft on ground
- Engines not running

Then:

Start levers in CUTOFF

↓

Engine fuel valves closed

↓

ENG VALVE CLOSED lights ON

👉 This is **completely normal**

⚠️ 2. Abnormal (if engines should be running)

If engines are supposed to be running:

⚠️ ENG VALVE CLOSED would mean:

- No fuel reaching engine
- Engine will shut down or fail to start

🧠 Line maintenance mindset

If ENG VALVE CLOSED + engines OFF:

✅ Normal indication

If ENG VALVE CLOSED + engines should be running:

👉 Check:

- Start lever position
- Fuel control system
- Electrical control of valve
- Possible fuel valve failure

🔗 Relationship with other lights in your image

Light	Meaning
ENG VALVE CLOSED	Fuel shutoff valve closed
SPAR VALVE CLOSED	Wing fuel shutoff valve closed
FILTER BYPASS	Fuel bypassing clogged filter
LOW PRESSURE	Fuel pumps not supplying pressure

⚡ Simple way to remember

ENG VALVE CLOSED = “engine is starved of fuel”

☑ Summary

- ● **ENG VALVE CLOSED (blue)** = engine fuel valve is closed
- Means:
 - No fuel to engine
- In your image: 👉 Most likely **normal because engines are shut down**

💡 Big-picture insight (tying everything together)

In your panel:

- Electrical system: **FAILED** ✅
- Fuel system: showing **low pressure & bypass** ✅
- Engine fuel valves: **CLOSED** ✅

👉 This strongly suggests:

- ✅ **Engines are shut down (which is why fuel valves are closed)**
- ✅ The rest of the panel is showing a **major electrical failure scenario**

💧 Final AMT takeaway

When you see:

- **ENG VALVE CLOSED (blue)**

👉 Always ask first:

"Are the engines running or shut down?"

✔ If shut down → **normal**

⚠ If running → **fuel supply problem**

FUEL LOW PRESSURE LIGHT ■ **FUEL LOW PRESSURE Light (Amber)** — What it means on the 737

Looking at your panel (fuel section, lower center-left of the image), you have:

- ✅ **LOW PRESSURE lights illuminated under the fuel pumps**
- Multiple pumps showing **LOW PRESSURE (amber)**
- Also nearby:

- **FILTER BYPASS lights (amber)**
- Engine/spare valves closed (blue)

Definition

The **FUEL LOW PRESSURE light** indicates that the **associated fuel pump is NOT delivering adequate fuel pressure**.

What the fuel pumps do

The 737 uses **electric boost pumps** in the tanks to:

- Deliver fuel **under pressure** to the engine
- Prevent fuel cavitation
- Ensure steady fuel flow

Normal operation

- Pump switch ON
- Pump running
-  No LOW PRESSURE light

What LOW PRESSURE means

 The pump is either:

- Not working
- Not powered
- Or cannot produce pressure

Common causes

1. **Pump not powered (VERY IMPORTANT in your case)**

- Electrical system failure
- Pump has no power → no pressure

 This matches your panel perfectly

2. **Pump failure**

- Motor failure
- Internal damage

3. **Low fuel level**

- Not enough fuel around pump inlet

4. **Air in the system / cavitation**

- Disrupted flow conditions

Your panel (CRITICAL interpretation)

You have:

-  **Multiple FUEL LOW PRESSURE lights ON**
-  **FILTER BYPASS lights ON (both engines)**
-  **ENG & SPAR VALVES CLOSED (blue)**
-  **Massive electrical failure indications (from rest of your panel):**
 - **DRIVE lights (both engines)**
 - **TRANSFER BUS OFF**
 - **SOURCE OFF**
 - **BAT DISCHARGE**
 - **TR UNIT**
 - **ELEC**
 - **STANDBY PWR OFF**

What's actually happening (step-by-step)

Both IDGs lost → DRIVE lights

↓

AC electrical power lost

↓

Fuel pumps lose electrical power

↓

Pumps stop running

↓

No fuel pressure generated

↓
FUEL LOW PRESSURE lights ON

✦ **Key insight (VERY important)**

⚠ **This is NOT a fuel pump failure problem**

👉 It is:

✅ **A result of electrical power loss**

🔗 **Tie to FILTER BYPASS lights**

This is important since both are ON:

Fuel pumps not running
↓
Fuel pressure drops
↓
Flow through filters becomes abnormal
↓
Filter restriction condition triggered
↓
FILTER BYPASS lights ON

🧠 **Line maintenance mindset**

If ONE LOW PRESSURE:

👉 Likely:

- Pump failure
- Electrical issue local to pump

If MULTIPLE LOW PRESSURE (your case):

👉 Immediately think:

✅ **Electrical power problem affecting all pumps**

Important relationships

Light	What it means
LOW PRESSURE	Pump not producing pressure
FILTER BYPASS	Fuel bypassing clogged/restricted filter
ENG VALVE CLOSED	Fuel valve closed (no flow)
ELEC / TR / DRIVE	Root cause → power loss

Simple way to remember

FUEL LOW PRESSURE = “pumps not supplying fuel pressure”

 In your case:

“...because electrical power is lost”

Summary (your exact panel)

-  Multiple **FUEL LOW PRESSURE** lights ON
-  Pumps are not operating
-  Electrical system is failed

 Final diagnosis:

 Fuel pumps are not running due to total electrical power loss — NOT individual pump failures

BIG PICTURE (everything you built)

Your entire aircraft scenario now clearly shows:

 Loss of both IDGs (DRIVE lights) → total AC power loss

- Transfer buses OFF
- TR units fail
- Battery discharges
- Fuel pumps lose power
- LOW PRESSURE lights
- Filter bypass conditions
- System-wide failures

Final AMT takeaway

When you see:

-  **Multiple LOW PRESSURE lights (fuel pumps)**
-  Along with **electrical failures**

 Always conclude:

 **Root cause = electrical power loss — not fuel system failure**

MAINT LIGHT

MAINT Light (Blue) — What it means on the 737

Looking at your image (bottom center of the electrical/APU section), you have:

-  **MAINT light (blue) illuminated**
- Nearby:
 - **LOW OIL PRESSURE (APU)**
 - **FAULT**
 - **OVERSPEED**
 - APU-related panel

Definition

The **MAINT light** indicates that a **maintenance fault has been detected and stored in the system (usually the APU or another monitored system)**.

What the MAINT light actually is

The MAINT light is **not a flight crew warning**.

 It is:

 **A maintenance advisory / stored fault indication**

It tells maintenance:

- “Hey, something happened — check the system”

Where it comes from

Most commonly on the 737:

-  **APU system (Primary source)**
- Sometimes:
 - Other monitored systems with BITE (Built-In Test Equipment)

What triggers the MAINT light

1. **APU fault recorded (MOST COMMON)**

- APU experienced a fault condition such as:
 - Start problems
 - Sensor issues
 - Minor system faults

2. **Stored fault in BITE system**

- Fault occurred earlier
- Stored for maintenance retrieval

3. **Electrical anomalies**

- Power interruptions can trigger APU/system faults

 Very relevant to your panel

Your panel (CRITICAL interpretation)

From your entire panel you've shown:

-  **DRIVE lights (both IDGs)**
-  **TRANSFER BUS OFF**
-  **SOURCE OFF**
-  **BAT DISCHARGE**
-  **TR UNIT**
-  **ELEC**
-  **STANDBY PWR OFF**
-  **EQUIP COOLING OFF**

🔑 What that means for MAINT:

Electrical system failure



APU and systems experience abnormal conditions



Faults recorded in system memory



MAINT light illuminates

★ Key insight (VERY important)

✅ MAINT light = “something is recorded for maintenance”

❌ It does NOT tell you the exact failure

👉 It is:

- A **symptom/flag**
- NOT the root cause

🧠 Line maintenance mindset

When you see MAINT:

👉 You DO NOT troubleshoot the light directly

Instead:

1. Access **BITE data**
2. Check:
 - a. APU fault codes
 - b. System fault history

In your case:

✅ MAINT is **secondary to the electrical failure**

🔑 Difference from other lights

Light	Meaning
-------	---------

MAINT (blue)	Maintenance advisory 
FAULT (amber)	Active failure
LOW OIL PRESS / OVERSPEED	Specific APU faults
ELEC	System-level electrical failure

⚡ Simple way to remember

MAINT = “maintenance needs to check something later”

☑ Summary

-  **MAINT light** = fault recorded in system memory
- Most common source:
 -  APU system
- In your panel:  Caused by **electrical system disruptions triggering stored faults**

💡 Big-picture (your full aircraft scenario)

You’ve now basically identified:

⚠ **Primary failure: Loss of both IDGs (DRIVE lights)**

- AC power lost
- Transfer buses OFF
- TR units fail
- Battery discharging
- Systems fail
- APU/system logs faults → **MAINT light ON**

💧 **Final AMT takeaway**

When you see:

- **MAINT light**

 Always think:

 “There’s a stored fault — go check BITE”

NOT:

✗ "The MAINT light itself is the problem"

APU LOW OIL PRESSURE LIGHT

▣ APU LOW OIL PRESSURE Light (Amber) — What it means on the 737

Looking at your image (bottom center APU panel), the **LOW OIL PRESSURE light (amber)** is illuminated next to the **MAINT, FAULT, and OVERSPEED lights**.

☑ Definition

The **APU LOW OIL PRESSURE light** indicates that the **engine oil pressure inside the APU has dropped below the safe operating limit**.

🔑 What the APU oil system does

The APU (Auxiliary Power Unit) uses oil to:

- ✓ Lubricate rotating components
- ✓ Reduce friction
- ✓ Remove heat
- ✓ Prevent damage to bearings and gears

🔔 What LOW OIL PRESSURE means

⚠ The APU is not properly lubricated

👉 This is **critical**, because:

- Moving parts can **overheat**
- Internal damage can occur **very quickly**

⚠ What causes LOW OIL PRESSURE

1. 💧 Low oil quantity (MOST COMMON)

- Insufficient oil in the APU reservoir
- Leak or improper servicing

2. Oil pump failure

- Pump not circulating oil properly

3. Excessive temperature

- Oil becomes too thin
- Pressure drops

4. Internal APU damage

- Bearing or internal component issues

System response (VERY important)

When LOW OIL PRESSURE is detected:

- ✓ The APU will **automatically shut down** to protect itself

Your panel (CRITICAL context)

From your full image:

- ✓ **LOW OIL PRESSURE light ON**
- ✓ **MAINT light ON**
- ✓ **FAULT light ON**
- ✓ **No AC electrical power available (from earlier):**
 - **DRIVE lights (both IDGs)**
 - **TRANSFER BUS OFF**
 - **SOURCE OFF**
 - **TR UNIT**
 - **BAT DISCHARGE**

How it fits into your scenario

 Likely sequence:

Major electrical failure (loss of IDGs)

↓

APU likely started or required for backup power



APU experiences abnormal condition



LOW OIL PRESSURE detected



APU shuts down



FAULT + MAINT lights come ON



No available AC power source remains

✦ BIG takeaway

⚠️ **APU failed or shut down due to low oil pressure**

👉 That means:

! **You lost your last backup electrical power source**

🧠 Line maintenance mindset

If LOW OIL PRESSURE alone:

👉 Check:

- Oil quantity
- Oil leaks
- Pump condition

If LOW OIL PRESSURE + FAULT + MAINT (your case):

👉 Strong indication:

✅ **APU experienced a serious fault and shut down**

🔑 Relationship with nearby lights

Light	Meaning
-------	---------

LOW OIL PRESSURE	APU lubrication failure 🚨
FAULT	APU shut down due to fault
MAINT	Fault stored in memory
OVERSPEED	APU exceeded speed limit (not active here)

⚡ Simple way to remember

LOW OIL PRESSURE = “APU is not being lubricated → will shut down”

☑ Summary (your exact panel)

- ✅ **APU LOW OIL PRESSURE (amber)** = oil pressure too low
- ✅ APU will **auto shut down**
- ✅ Combined with:
 - FAULT
 - MAINT

👉 Conclusion:

⚠ **APU shut down due to oil system failure**

💡 **BIG FINAL PICTURE (everything you’ve shown)**

You now have a complete chain:

⚠ **Both IDGs lost (DRIVE lights)**

- No AC from engines
- APU needed as backup
- APU fails (LOW OIL PRESSURE)
- No AC source available
- SOURCE OFF + TRANSFER BUS OFF
- TR units fail
- Battery discharges
- System-wide aircraft failures

💧 **Final AMT takeaway**

When you see:

- **DRIVE lights (both engines)**
- **APU LOW OIL PRESSURE + FAULT**
- **SOURCE OFF + TRANSFER BUS OFF**

👉 You can confidently diagnose:

✅ **Complete loss of ALL electrical generation (engine generators + APU)**

APU FAULT LIGHT

📺 **APU LOW OIL PRESSURE Light (Amber) — What it means on the 737**

Looking at your image (bottom center APU panel), the **LOW OIL PRESSURE light (amber)** is illuminated next to the **MAINT, FAULT, and OVERSPEED lights**.

✅ **Definition**

The **APU LOW OIL PRESSURE light** indicates that the **engine oil pressure inside the APU has dropped below the safe operating limit**.

🔑 **What the APU oil system does**

The APU (Auxiliary Power Unit) uses oil to:

- ✅ Lubricate rotating components
- ✅ Reduce friction
- ✅ Remove heat
- ✅ Prevent damage to bearings and gears

🔔 **What LOW OIL PRESSURE means**

⚠️ **The APU is not properly lubricated**

👉 This is **critical**, because:

- Moving parts can **overheat**
- Internal damage can occur **very quickly**

What causes LOW OIL PRESSURE

1. Low oil quantity (MOST COMMON)

- Insufficient oil in the APU reservoir
- Leak or improper servicing

2. Oil pump failure

- Pump not circulating oil properly

3. Excessive temperature

- Oil becomes too thin
- Pressure drops

4. Internal APU damage

- Bearing or internal component issues

System response (VERY important)

When LOW OIL PRESSURE is detected:

- ✓ The APU will **automatically shut down** to protect itself

Your panel (CRITICAL context)

From your full image:

- ✓ **LOW OIL PRESSURE light ON**
- ✓ **MAINT light ON**
- ✓ **FAULT light ON**
- ✓ **No AC electrical power available (from earlier):**
 - **DRIVE lights (both IDGs)**
 - **TRANSFER BUS OFF**
 - **SOURCE OFF**
 - **TR UNIT**
 - **BAT DISCHARGE**

How it fits into your scenario

 Likely sequence:

Major electrical failure (loss of IDGs)
↓
APU likely started or required for backup power
↓
APU experiences abnormal condition
↓
LOW OIL PRESSURE detected
↓
APU shuts down
↓
FAULT + MAINT lights come ON
↓
No available AC power source remains

BIG takeaway

 **APU failed or shut down due to low oil pressure**

 That means:

 **You lost your last backup electrical power source**

Line maintenance mindset

If LOW OIL PRESSURE alone:

 Check:

- Oil quantity
- Oil leaks
- Pump condition

If LOW OIL PRESSURE + FAULT + MAINT (your case):

 Strong indication:

✅ **APU experienced a serious fault and shut down**

🔑 Relationship with nearby lights

Light	Meaning
LOW OIL PRESSURE	APU lubrication failure 🚨
FAULT	APU shut down due to fault
MAINT	Fault stored in memory
OVERSPEED	APU exceeded speed limit (not active here)

⚡ Simple way to remember

LOW OIL PRESSURE = “APU is not being lubricated → will shut down”

☑ Summary (your exact panel)

- ✅ **APU LOW OIL PRESSURE (amber)** = oil pressure too low
- ✅ APU will **auto shut down**
- ✅ Combined with:
 - FAULT
 - MAINT

👉 Conclusion:

⚠ **APU shut down due to oil system failure**

💡 **BIG FINAL PICTURE (everything you’ve shown)**

You now have a complete chain:

⚠ **Both IDGs lost (DRIVE lights)**

- No AC from engines
- APU needed as backup
- APU fails (LOW OIL PRESSURE)
- No AC source available
- SOURCE OFF + TRANSFER BUS OFF
- TR units fail

- Battery discharges
- System-wide aircraft failures

Final AMT takeaway

When you see:

- **DRIVE lights (both engines)**
- **APU LOW OIL PRESSURE + FAULT**
- **SOURCE OFF + TRANSFER BUS OFF**

👉 You can confidently diagnose:

- ✅ **Complete loss of ALL electrical generation (engine generators + APU)**

OVERSPEED

APU OVERSPEED Light (Amber) — What it means on the 737

Looking at your panel, the **OVERSPEED light (amber)** is located in the **APU section** (bottom center), alongside:

- ✅ **LOW OIL PRESSURE**
- ✅ **FAULT**
- ✅ **MAINT**

Definition

The **APU OVERSPEED light** indicates that the **APU has exceeded its maximum allowable rotational speed limit**.

What “overspeed” means

The APU is a small turbine engine that must stay within a **safe RPM range**.

⚠️ **Overspeed = the APU is spinning faster than design limits**

Why this is critical

If the APU overspeeds:

- Internal components can:
 - Fail
 - Break apart
- Serious damage can occur very quickly

System response (VERY important)

When overspeed is detected:

- ✓ The APU will **immediately shut itself down automatically**

And:

- ✓ **OVERSPEED light comes ON**
- ✓ **FAULT light also comes ON**
- ✓ **MAINT light records the event**

What causes OVERSPEED

1. **Fuel control malfunction**

- Too much fuel → excessive acceleration

2. **Governor/control failure**

- Speed control system fails to regulate RPM

3. **Load rejection**

- Sudden loss of electrical load
→ APU speeds up too quickly

4. **Electrical issues (VERY relevant to your panel)**

- Unstable electrical system
- Improper generator loading/unloading

🔗 Your panel (CRITICAL interpretation)

From your full panel:

- ☒ **LOW OIL PRESSURE**
- ☒ **FAULT**
- ☒ **MAINT**
- ☒ **OVERSPEED** (this indicator shown on panel)
- ☒ **Major electrical failure** (DRIVE lights, TRANSFER BUS OFF, SOURCE OFF, etc.)

🔗 Putting it together (your exact scenario)

You mainly have:

- ☒ **LOW OIL PRESSURE** (actual active fault)
- ☒ **FAULT + MAINT** (result of shutdown)

👉 The **OVERSPEED** light is present on the panel but **NOT illuminated** in your image.

👉 That means:

⚠️ **Your APU shutdown was NOT caused by overspeed**

Instead:

LOW OIL PRESSURE occurs
↓
APU lubrication unsafe
↓
APU shuts down
↓
FAULT + MAINT lights ON

🧠 Line maintenance mindset

If OVERSPEED light ON:

👉 Root cause:

- Speed control failure
- Fuel/governor issue

If OVERSPEED light OFF (your case):

👉 It is:

✅ **Not the cause of the failure**

🔑 Important relationships

Light	Role
LOW OIL PRESSURE	Root cause in your panel ✅
OVERSPEED	Alternative shutdown cause (not active)
FAULT	APU shut down
MAINT	Fault recorded

⚡ Simple way to remember

OVERSPEED = "APU spun too fast → immediate shutdown"

☑ Summary

- **OVERSPEED (amber)** = APU exceeded speed limit
- Causes:
 - Fuel control issue
 - Governor failure
 - Electrical/load instability
- In your image: 👉 **NOT illuminated** → **not the cause**

💡 BIG FINAL PICTURE (everything together)

Your full scenario is:

⚠ **Both IDGs lost (DRIVE lights)**

- No AC power
- APU required as backup
- APU runs into **LOW OIL PRESSURE**
- APU shuts down (**FAULT**)
- NO electrical source remains

- **SOURCE OFF + TRANSFER BUS OFF**
- Whole aircraft electrical system collapses

Final AMT takeaway

When analyzing APU lights:

- ✓ **The illuminated amber light = root cause**
- ✓ **FAULT = result**
- ✓ **MAINT = recorded event**